Predictive Thermal Battery Management System for Li-ion Battery Packs

Build guide: circuits, single-line diagram, wiring, component placement, assembly, and the simulation in the project simulator, KiCad and GNU Octave

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Design basis: architecture revision 2.2 (24 October 2026)
Built 30 October 2026 from the predictive_bms repository

The prototype is built from off-the-shelf modules on two perfboards; no PCB is made. Every simulator result in this guide comes from an illustrative cell model, not from the project's cells: the real values are measured at the build gates named beside them.

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1. How to use this guide

- Build in the order of section 12, and follow the gates of the safe build sequence (section 14). No lithium cell is connected before Gate 3.
- Holes are named by column and row, **(c, r)**, counted from 0 at the top-left corner of the component side. Column 0 is on the left; on the underside drawings it is on the right, because the board is turned over.
- Wire colours in the drawings mark nets, not the colour of wire to buy: red 3V3, orange 5 V, black GND, blue SDA, amber SCL, purple taps, green ADC inputs, teal NTC lines.
- The board layouts are generated from one description of each board, which is checked automatically: every net is joined, and no two nets touch. The placement tables, wiring lists and continuity checks come from the same description.
- Generic modules differ. Before soldering a module's header, compare its printed pin names with section 10.

2. The system at a glance

Two independent layers protect the pack. **Layer A** is a hardware BMS that always limits cell voltage and current and balances the cells. **Layer B** is the ESP32 supervisor: five times a second it measures every cell voltage, the pack current, four cell temperatures and the air temperature, runs a two-state thermal model of every cell with a Kalman filter, and opens separate charge and load relays, latched, before a fixed limit is reached.

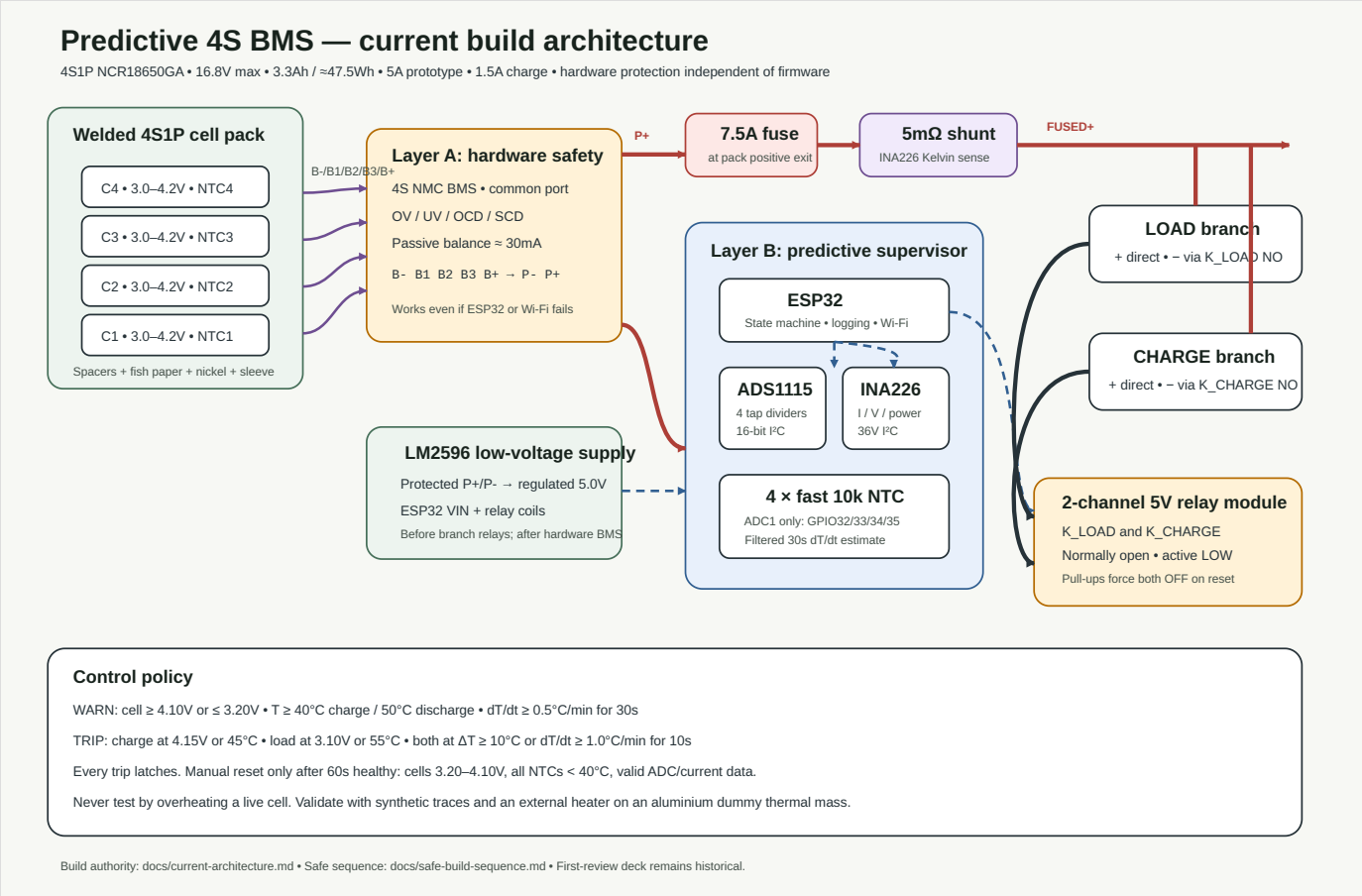


Figure 1. Architecture revision 2.2: power, protection and sensing.

Quantity	Value
Pack	4S1P NCR18650GA, 14.4 V nominal, 16.8 V max, 3.3 Ah (label)
Continuous discharge	5 A (software trip at 5.0 A); fuse 7.5 A at B+
Charge	isolated 16.8 V CC/CV, 1.5 A max (software trip at 1.6 A)
Cell limits (software)	warn 4.10 V / 3.20 V; trip 4.15 V (charge off) / 3.10 V (load off)
Temperature limits	charging 40 / 45 °C; discharging 50 / 55 °C; spread 7 / 10 °C
Predictive levels	abnormal heat 0.5 °C/min Warning, 1.0 °C/min Trip; cell difference 0.3 / 0.6 °C/min
Hardware BMS	OV 4.25 V, UV 2.7 V, short circuit, balance from 4.18 V at 30 mA

How the supervisor decides

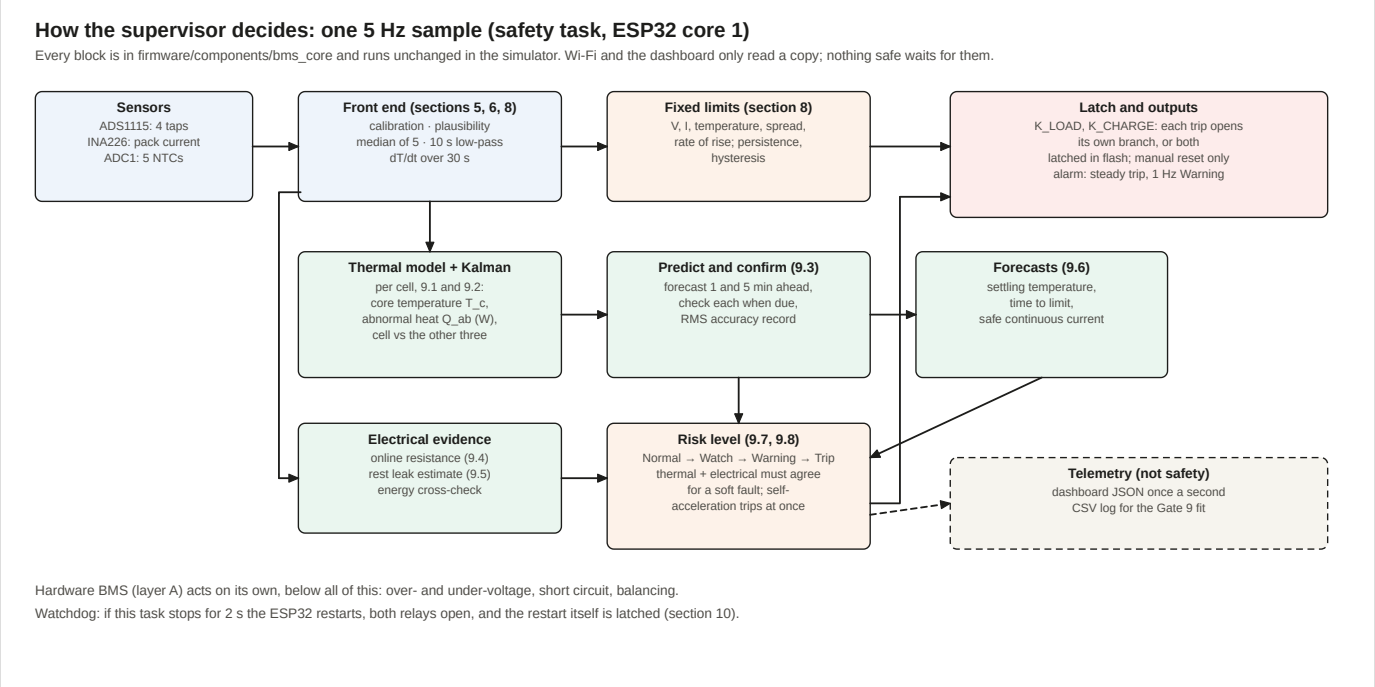


Figure 1a. One 5 Hz sample of the safety task, from the sensors to the relays.

Level	Entered when any one of these holds	Action
0 Normal	none of the conditions below	none
1 Watch	*thermal:* 5-minute forecast too low by 1.0 °C or more; abnormal heat 0.3 °C/min above the other cells for 30 s; under 10 min to the limit, or current above the safe current for 30 s. *Electrical:* leak at rest of 20 mA or more; resistance more than 30 % from its Gate 4 value	dashboard notice and log
2 Warning	abnormal heat of 0.5 °C/min (about 0.4 W) for 30 s, or a thermal and an electrical Watch on the same cell	alarm; both branches stay on
3 Trip	abnormal heat of 1.0 °C/min (about 0.75 W) for 10 s; 0.6 °C/min above the other cells for 10 s; self-accelerating heat while at Warning	both relays open, latched, logged

Risk levels of architecture section 9.8. They are fixed rules, not a trained classifier, so every level can be traced to the measurement that set it.

3. Single-line diagram

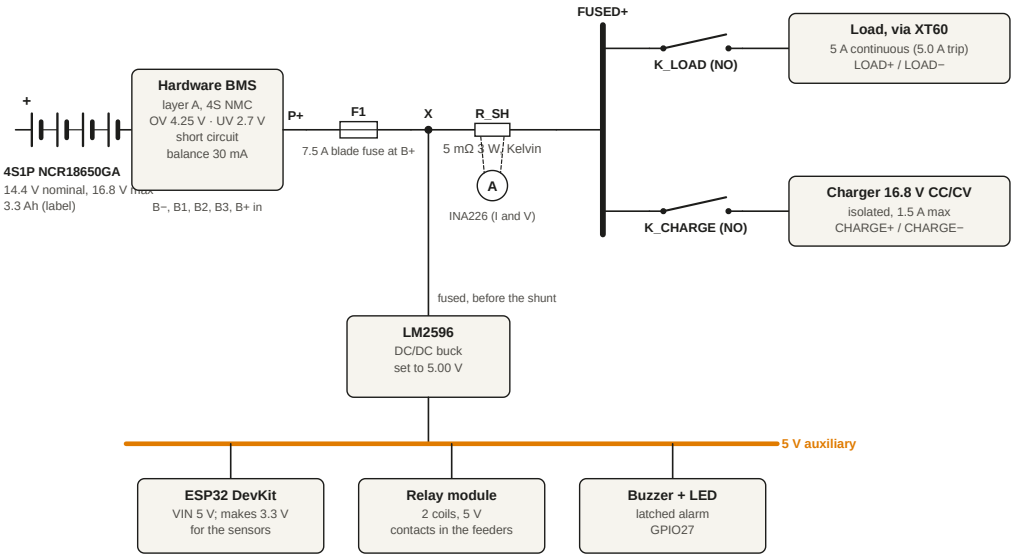


Figure 2. Single-line diagram of the power path and the 5 V auxiliary supply.

- The fuse sits at the pack's positive exit, then the shunt, then the branch wiring (section 11).
- K_LOAD and K_CHARGE are normally-open contacts. Pull-ups hold both open whenever the ESP32 is reset, crashed or unpowered.
- The LM2596 is fed after the fuse and before the shunt. Its own current therefore does not count as pack current, so the pack can read "at rest" (below 20 mA), which the self-discharge estimate needs.

4. Circuit diagrams

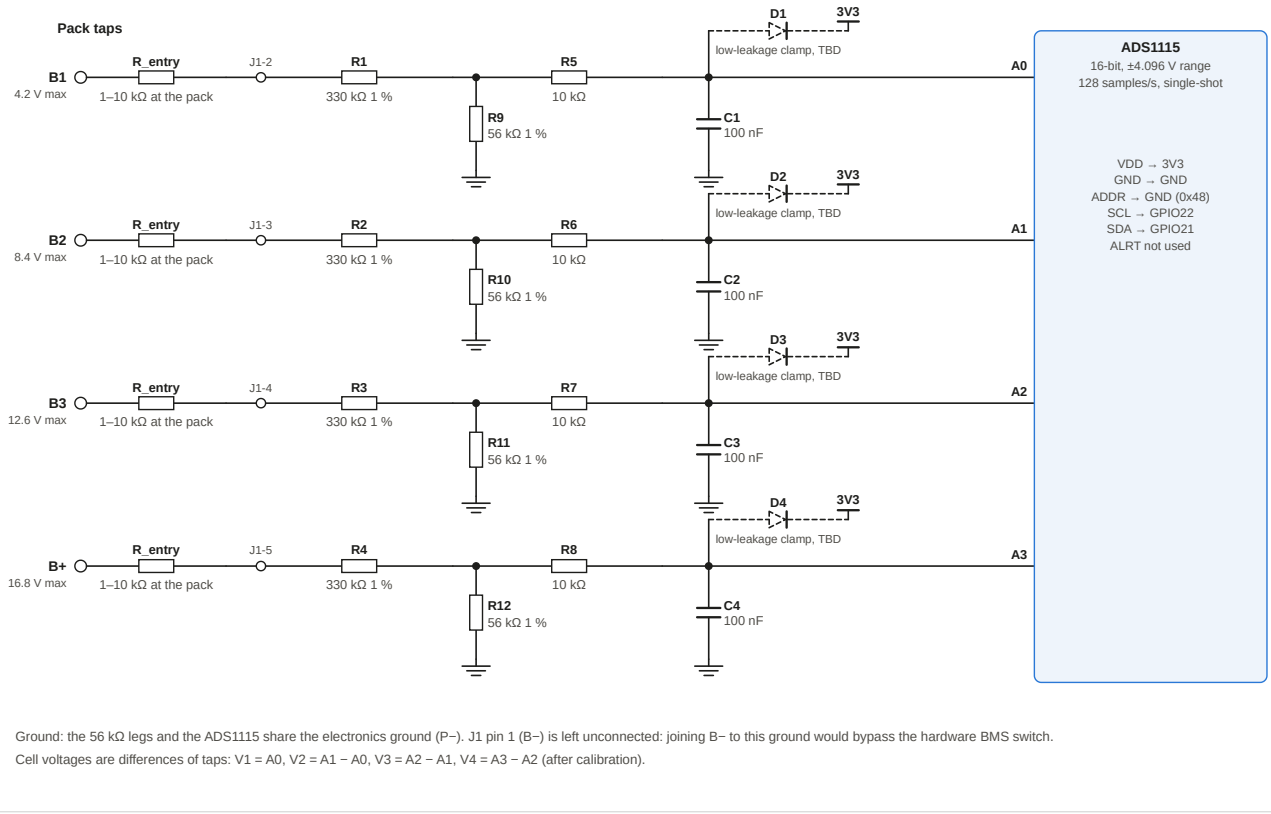
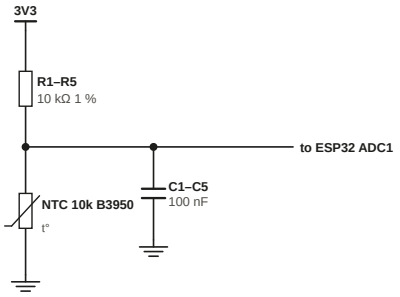


Figure 3. Cell-tap voltage sensing on board A.

Circuit: temperatures and current (board B, architecture sections 4 and 6)

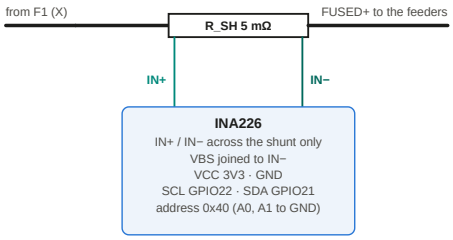
Five NTC dividers (NTC1–4 on the cells, NTC5 in the air)



Sensor	Where	Pin	Channel
NTC1	cell 1	GPIO32	ADC1_CH4
NTC2	cell 2	GPIO33	ADC1_CH5
NTC3	cell 3	GPIO34	ADC1_CH6
NTC4	cell 4	GPIO35	ADC1_CH7
NTC5	air	GPIO36 (VP)	ADC1_CH0

ADC1 only: ADC2 does not work while Wi-Fi is on. At 25 °C the node sits at 1.65 V.
Bond NTC1–4 at mid-can with insulating, thermally conductive tape; never on nickel.

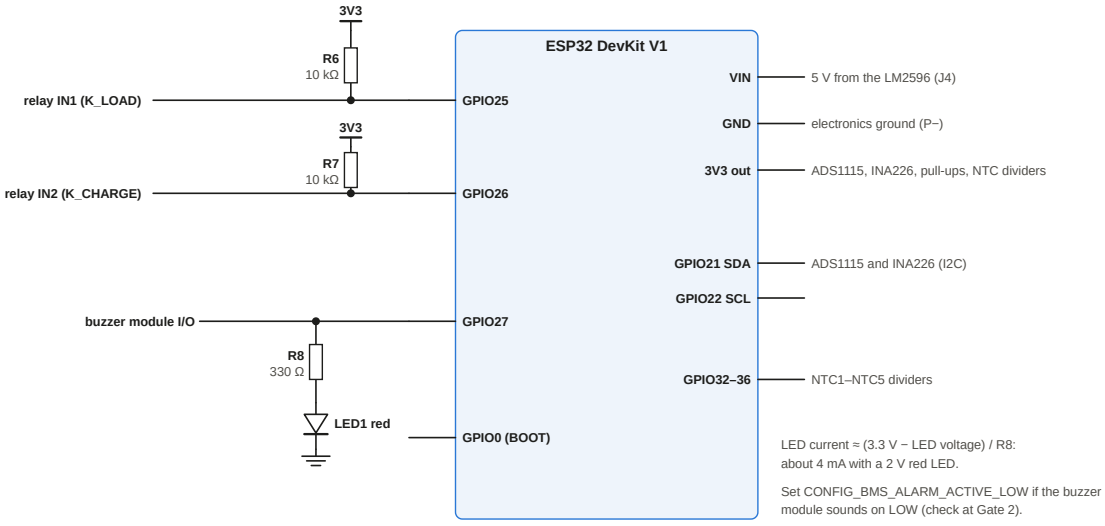
Pack current: 5 mΩ shunt with Kelvin sense, INA226



- Remove or bypass the module's own R100 shunt (BOM item 12).
- Twist the two sense leads; take them from the shunt's own pads (Kelvin).
- At 5 A the shunt drops 25 mV and dissipates 0.125 W.
- Positive current = discharge (IN+ on the fuse side).

Figure 4. Temperature dividers and the current measurement on board B.

Circuit: ESP32 control, relays and alarm (board B, architecture section 7)



LED current ≈ (3.3 V − LED voltage) / R8:
about 4 mA with a 2 V red LED.

Set CONFIG_BMS_ALARM_ACTIVE_LOW if the buzzer module sounds on LOW (check at Gate 2).

Relay module
2 × 5 V inputs, active LOW
VCC 5 V · GND

LOW closes a contact. HIGH or floating leaves it open, so reset, a crash or an unpowered ESP32 leave both branches open.

Steady alarm = latched trip. Pulsing at 1 Hz = risk level Warning.

Check at Gate 2 that a 3.3 V HIGH fully releases each 5 V relay input; if not, add a transistor stage (section 7).

Figure 5. ESP32 control: relay inputs with pull-ups, alarm buzzer and LED.

5. ESP32 pin map

Function	ESP32 pin	Board B	Notes
I2C SDA	GPIO21	J7, J8	ADS1115 (0x48) and INA226 (0x40)
I2C SCL	GPIO22	J7, J8	4.7 kΩ pull-ups only if the modules have none
K_LOAD	GPIO25	J5 IN1	active LOW, 10 kΩ pull-up to 3V3
K_CHARGE	GPIO26	J5 IN2	active LOW, 10 kΩ pull-up to 3V3
Alarm buzzer and LED	GPIO27	J6 I/O, LED1	steady = trip, 1 Hz = Warning
NTC1 (cell 1)	GPIO32 · ADC1_CH4	J9	ADC1 only
NTC2 (cell 2)	GPIO33 · ADC1_CH5	J10	
NTC3 (cell 3)	GPIO34 · ADC1_CH6	J11	input-only pin
NTC4 (cell 4)	GPIO35 · ADC1_CH7	J12	input-only pin
NTC5 (air)	GPIO36 (VP) · ADC1_CH0	J13	input-only pin
Manual reset	GPIO0 (BOOT)	on the DevKit	hold 3 s; or `reset` on the console
Power	VIN 5 V, 3V3 out	J4, rails	VIN from the LM2596 set to 5.00 V

6. Components

6.1 Bill of materials (architecture BOM)

Full replacement value of the active items: **₹9,260.19** (prices from `bom.csv`; shipping excluded; items marked "existing stock" are owned already).

#	Part	Specification	Qty	Line total	Status
1	Microcontroller	ESP32-WROOM-32 DevKit v1 30-pin	1 pcs	₹639.00	In cart
2	Li-ion cells	Panasonic/Sanyo NCR18650GA 3.6V 3300mAh cells; matched 4S1P set	4 pcs	₹3,596.00	Available from existing stock
3	Independent hardware BMS	4S NMC common-port board; 20A hardware rating; OV/UV/OCD/SCD and passive balance	1 pcs	₹239.00	In cart
4	Cell spacers	18650 2x1 modular spacers; upper and lower layers	4 pcs	₹16.00	In cart
5	Positive terminal rings	18650 fish-paper adhesive terminal insulation 20-hole sheet	2 sheet	₹12.00	In cart
6	Barrier paper and pack sleeve	Fish/barley paper plus appropriately sized PVC heat-shrink sleeve	1 set	₹150.00	Verify before order
7	Pure nickel strip	0.15x8mm pure nickel strip	1 pack	₹569.00	Available from existing stock
8	Fast NTC thermistors	10k B3950 epoxy bead thermistors	5 pcs	₹70.00	Add one to cart
9	External ADC	ADS1115 16-bit 4-channel I2C ADC at 3.3V	1 pcs	₹119.00	In cart
10	Precision resistor kit	1% metal-film values including 10k; 56k; 330k	1 set	₹389.00	In cart
11	RC filters and ADC clamps	100nF capacitors; 10k series resistors; low-leakage 3.3V clamps	1 set	₹100.00	Verify before order
12	Current and power monitor	INA226 36V bidirectional I2C module	1 pcs	₹124.00	In cart

#	Part	Specification	Qty	Line total	Status
13	External shunt	5mOhm 3W 1% Kelvin-connected shunt pack of 3	1 pack	₹15.00	In cart
14	Buck regulator	LM2596 adjustable buck set to 5.0V before connection	1 pcs	₹45.00	In cart
15	Charge and load disconnect	Dual 5V relay module; 2 independently controlled NO contacts; 10A at 30VDC	1 pcs	₹209.00	In cart
16	Pack fuse	7.5A 32V medium ATC/ATO blade fuse	2 pcs	₹22.00	In cart
17	Inline fuse holder	Waterproof inline blade-fuse holder pack of 5	1 pack	₹589.00	In cart
18	Power and balance connectors	2 XT60 pairs plus 5-pin JST-XH balance or tap harness	1 set	₹140.00	In cart
19	Power and sense wire	16AWG red/black power wire plus five colours of 22AWG sense wire	7 m	₹198.00	In cart
20	Perfboard	FR4 7x9cm 2.54mm single-sided prototyping board	2 pcs	₹78.19	Available from existing stock
21	Dedicated charger	Isolated 16.8V CC/CV charger; 1.5A maximum; for 4S NMC or Li-ion	1 pcs	₹850.00	Available from existing stock
22	Charger connector	Barrel adapter pair matched to the selected charger's measured plug; provisional 5.5x2.5mm	1 pair	₹169.00	Hold - verify charger plug
23	PETG filament	1.75mm PETG light grey 1kg spool	1 spool	₹780.00	Available from existing stock
24	Mounting and strain relief	M3 standoffs; screws; cable glands; ferrules; ties and adhesive mounts	1 set	₹100.00	Verify before order
25	Alarm	5V active buzzer module plus red status LED and resistor	1 set	₹42.00	In cart

6.2 Passive parts on the two boards

Part	Value	Qty	Source
Ceramic capacitor	100n	9	BOM item 11
Clamp diode	clamp (TBD)	4	BOM item 11: chosen after the Gate 1 leakage test
LED	red LED	1	stock (BOM item 25)
Resistor 1/4 W 1 % metal film	10k	11	kit (BOM item 10)
Resistor 1/4 W 1 % metal film	330R	1	kit; any value that gives a few mA in the LED
Resistor 1/4 W 1 % metal film	330k	4	kit (BOM item 10): check the value is in it
Resistor 1/4 W 1 % metal film	56k	4	kit (BOM item 10): check the value is in it
Resistor 1/4 W (tap entry, at the pack end)	1–10 kΩ, e.g. 10k	4	kit (section 11)

6.3 Assembly hardware to add to the order

These small parts are needed to build the boards as drawn and are not lines of the BOM yet.

Part	Qty	Used for
Female header 1 × 15, 2.54 mm	2	ESP32 DevKit sockets (board B, U1)
Female header 1 × 10, 2.54 mm	1	ADS1115 socket (board A, J2)
Female header 1 × 8, 2.54 mm	1	INA226 socket (board B, J8)

Part	Qty	Used for
Male header strip, 2.54 mm	1 strip (40 pins)	J3, J7 (4 each), J5 (4), J6 (3), J14 (2), and spares
JST-XH 2-pin plug and socket with lead	5	NTC1–NTC5 (board B, J9–J13)
JST-XH 5-pin plug and socket with lead	1	tap harness (board A, J1); BOM item 18 has 4
Screw terminal, 2-pin, 5.08 mm	1	5 V in (board B, J4)
Female–female jumper leads, 20 cm	1 set (about 15)	board A ↔ B link, relay inputs, buzzer
Tinned copper wire, 0.5–0.6 mm (or resistor offcuts)	about 1 m	bare runs and bridges on both boards
Insulated hook-up wire, 24–30 AWG	about 2 m	the insulated links (35 on both boards)
Heat-shrink, 2–3 mm	1 m	tap-entry resistors and NTC leads
M3 nylon standoffs and screws	8	both boards (part of BOM item 24)

7. Wiring between the units

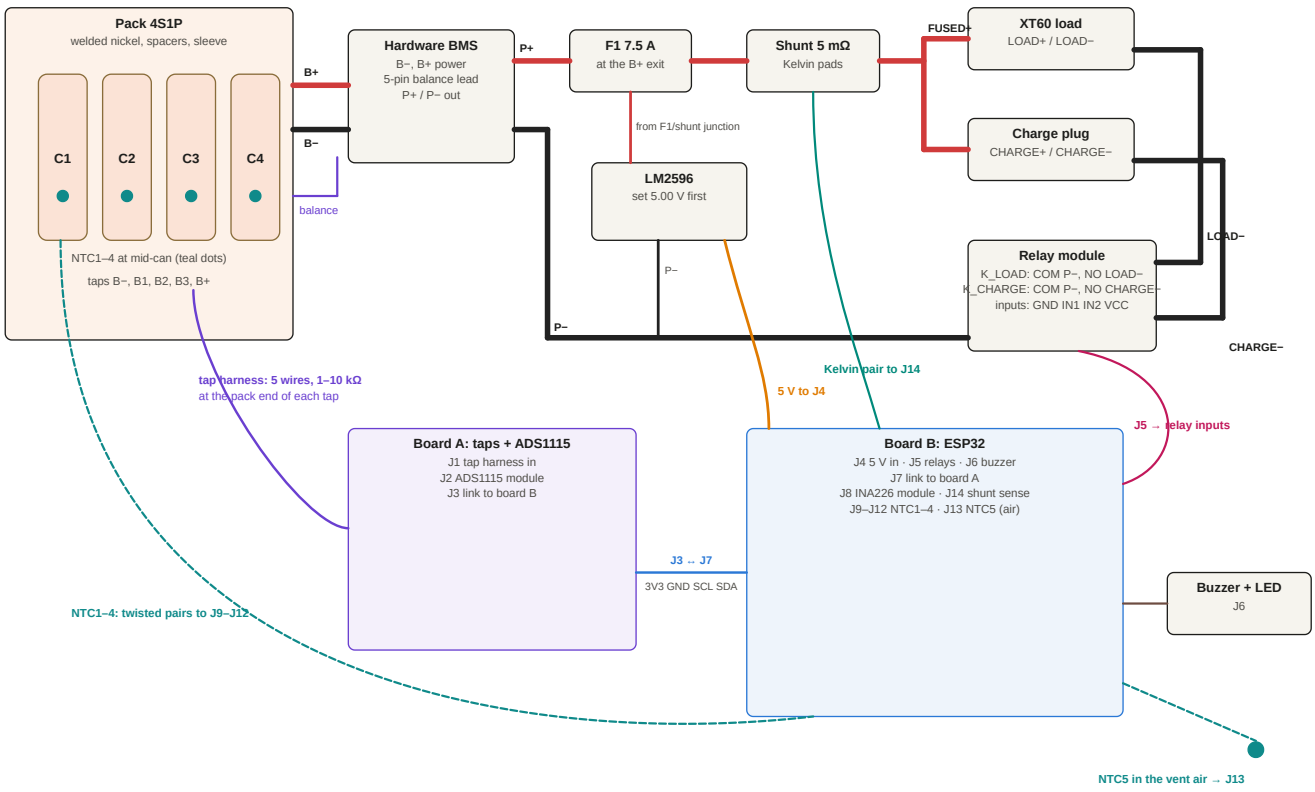


Figure 6. Every connection between the pack, the power parts, the two boards and the modules.

Connectors

Connector	Goes to
J1 B-	nothing (see the note on the sense ground)
J1 B1, B2, B3, B+	the pack's taps, each through a 1-10 kΩ resistor at the pack end
J2	ADS1115 module (plugs in)
J3	board B J7: 3V3, GND, SCL, SDA
J4	LM2596 output, set to 5.00 V before connecting
J5 GND, IN1, IN2, VCC	relay module GND, IN1 (K_LOAD), IN2 (K_CHARGE), VCC
J6 VCC, I/O, GND	active buzzer module
J7	board A J3: 3V3, GND, SCL, SDA
J8	INA226 module (plugs in)
J9-J12	NTC1-NTC4 on cells 1-4
J13	NTC5 in the air of the vent path
J14 IN+, IN-	the shunt's Kelvin sense pair: IN+ on the fuse side, IN- on the load side

> Tap harness: fit a 1–10 kΩ resistor at the pack end of each tap wire (B1, B2, B3, B+), sleeved in heat-shrink, so a damaged wire cannot short a cell. Never hot-plug a harness that has not been measured: each adjacent pair must read one cell voltage, rising in order. Connect B− first.

8. Board A: cell-tap sensing

Four identical divider channels feed the ADS1115. Parts are placed by channel, left to right in the direction of the signal: tap in, 330 kΩ, the node, 10 kΩ and 56 kΩ, then the filter capacitor and the clamp. Two bare buses run down the board: GND at column 15 and 3V3 at column 17.

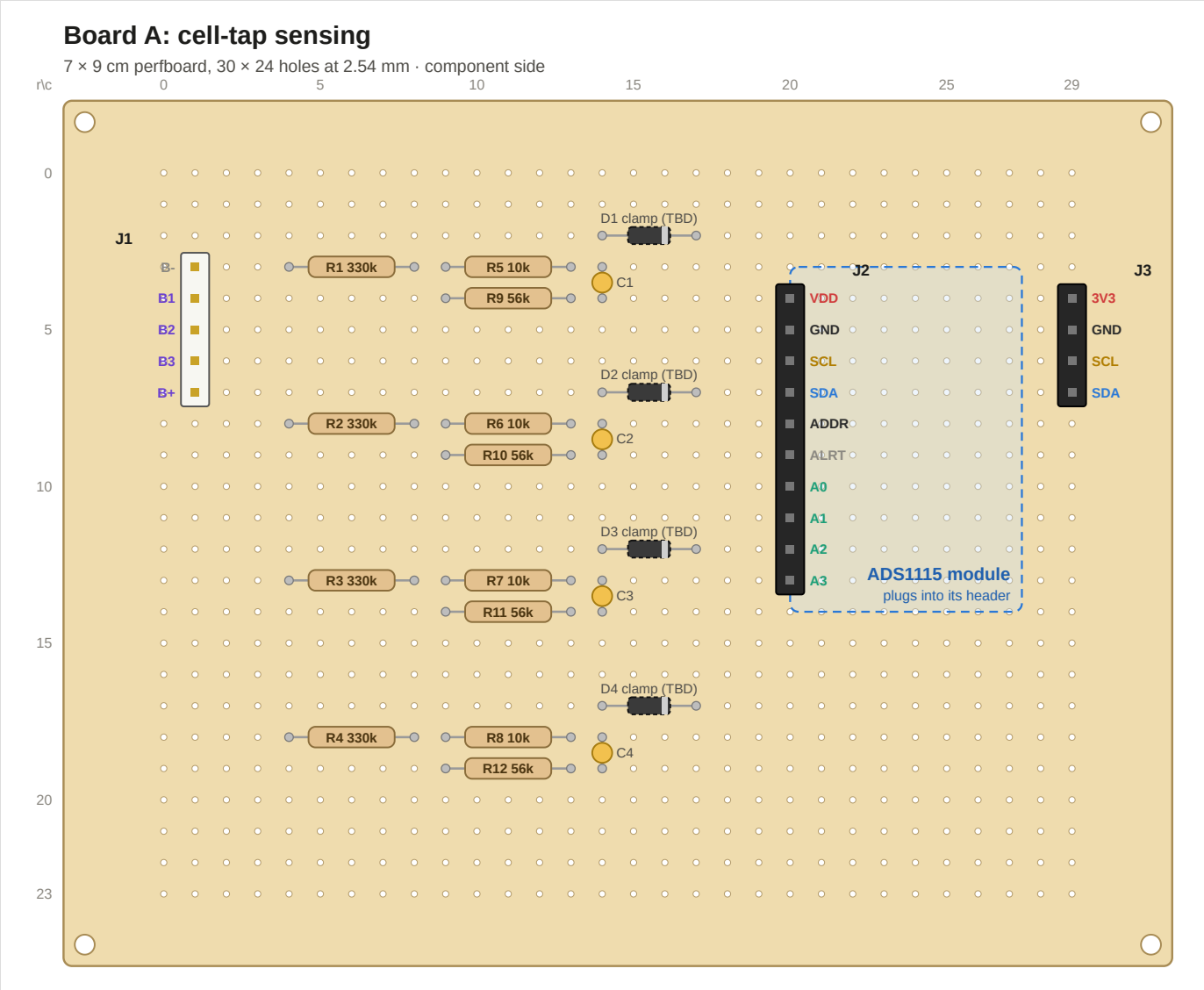


Figure 7. Board A, component side.

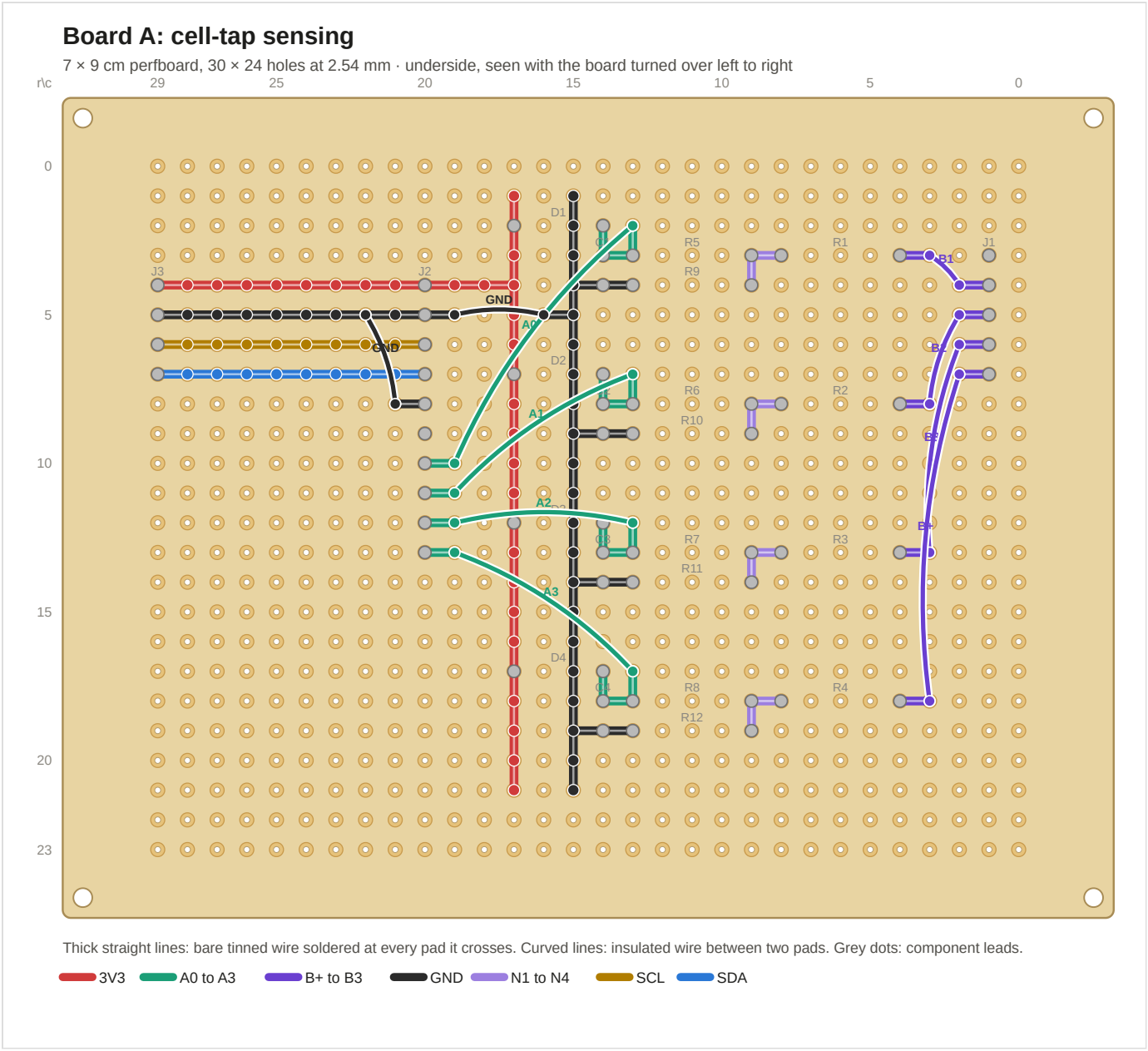


Figure 8. Board A, underside (turned over left to right).

> J1 pin 1 (B−) is deliberately not connected. Joining B− to this board's ground would join B− and P− through a thin wire and bypass the hardware BMS's switch.

Board A: Pin-to-pin wiring list (schematic connections)

If you did not use the exact hole coordinates of the drawing, use this list. Every pin listed in a row connects to the other pins in that same row:

Net	Pins to join together	Function
3V3	**D1** pin K (clamp (TBD)) · **D2** pin K (clamp (TBD)) · **D3** pin K (clamp (TBD)) · **D4** pin K (clamp (TBD)) · **ADS1115** VDD · **Board link J3** pin 3V3	3.3 V logic & sensor supply bus
A0	**R5** pin 2 (10k) · **C1** pin 1 (100n) · **D1** pin A (clamp (TBD)) · **ADS1115** A0	Filtered channel 1 voltage to ADS1115 input A0
A1	**R6** pin 2 (10k) · **C2** pin 1 (100n) · **D2** pin A (clamp (TBD)) · **ADS1115** A1	Filtered channel 2 voltage to ADS1115 input A1
A2	**R7** pin 2 (10k) · **C3** pin 1 (100n) · **D3** pin A (clamp (TBD)) · **ADS1115** A2	Filtered channel 3 voltage to ADS1115 input A2
A3	**R8** pin 2 (10k) · **C4** pin 1 (100n) · **D4** pin A (clamp (TBD)) · **ADS1115** A3	Filtered channel 4 voltage to ADS1115 input A3
B+	**Tap connector J1** pin B+ · **R4** pin 1 (330k)	Cell 4 top tap from pack (+16.8 V max) to 330 kΩ divider
B1	**Tap connector J1** pin B1 · **R1** pin 1 (330k)	Cell 1 tap from pack (+4.2 V max) to 330 kΩ divider
B2	**Tap connector J1** pin B2 · **R2** pin 1 (330k)	Cell 2 tap from pack (+8.4 V max) to 330 kΩ divider
B3	**Tap connector J1** pin B3 · **R3** pin 1 (330k)	Cell 3 tap from pack (+12.6 V max) to 330 kΩ divider
GND	**R9** pin 2 (56k) · **C1** pin 2 (100n) · **R10** pin 2 (56k) · **C2** pin 2 (100n) · **R11** pin 2 (56k) · **C3** pin 2 (100n) · **R12** pin 2 (56k) · **C4** pin 2 (100n) · **ADS1115** GND · **ADS1115** ADDR · **Board link J3** pin GND	Electronics ground bus (P−)
N1	**R1** pin 2 (330k) · **R5** pin 1 (10k) · **R9** pin 1 (56k)	Channel 1 divider junction (330k, 56k to GND, 10k series)
N2	**R2** pin 2 (330k) · **R6** pin 1 (10k) · **R10** pin 1 (56k)	Channel 2 divider junction (330k, 56k to GND, 10k series)
N3	**R3** pin 2 (330k) · **R7** pin 1 (10k) · **R11** pin 1 (56k)	Channel 3 divider junction (330k, 56k to GND, 10k series)
N4	**R4** pin 2 (330k) · **R8** pin 1 (10k) · **R12** pin 1 (56k)	Channel 4 divider junction (330k, 56k to GND, 10k series)
SCL	**ADS1115** SCL · **Board link J3** pin SCL	I2C clock line (ESP32 GPIO22 to ADS1115 & INA226)
SDA	**ADS1115** SDA · **Board link J3** pin SDA	I2C data line (ESP32 GPIO21 to ADS1115 & INA226)

Board A: placement

Ref	Part	Pins and holes (c, r)	Note
J1	JST-XH 5-pin	B- (1, 3) · B1 (1, 4) · B2 (1, 5) · B3 (1, 6) · B+ (1, 7)	tap harness from the pack
R1	330k	1 (4, 3) · 2 (8, 3)	
R5	10k	1 (9, 3) · 2 (13, 3)	

Ref	Part	Pins and holes (c, r)	Note
R9	56k	1 (9, 4) · 2 (13, 4)	
C1	100n	1 (14, 3) · 2 (14, 4)	
D1	clamp (TBD)	A (14, 2) · K (17, 2)	fit after the Gate 1 leakage test
R2	330k	1 (4, 8) · 2 (8, 8)	
R6	10k	1 (9, 8) · 2 (13, 8)	
R10	56k	1 (9, 9) · 2 (13, 9)	
C2	100n	1 (14, 8) · 2 (14, 9)	
D2	clamp (TBD)	A (14, 7) · K (17, 7)	fit after the Gate 1 leakage test
R3	330k	1 (4, 13) · 2 (8, 13)	
R7	10k	1 (9, 13) · 2 (13, 13)	
R11	56k	1 (9, 14) · 2 (13, 14)	
C3	100n	1 (14, 13) · 2 (14, 14)	
D3	clamp (TBD)	A (14, 12) · K (17, 12)	fit after the Gate 1 leakage test
R4	330k	1 (4, 18) · 2 (8, 18)	
R8	10k	1 (9, 18) · 2 (13, 18)	
R12	56k	1 (9, 19) · 2 (13, 19)	
C4	100n	1 (14, 18) · 2 (14, 19)	
D4	clamp (TBD)	A (14, 17) · K (17, 17)	fit after the Gate 1 leakage test
J2	ADS1115 module	VDD (20, 4) · GND (20, 5) · SCL (20, 6) · SDA (20, 7) · ADDR (20, 8) · ALRT (20, 9) · A0 (20, 10) · A1 (20, 11) · A2 (20, 12) · A3 (20, 13)	1 x 10 female header
J3	4-pin header	3V3 (29, 4) · GND (29, 5) · SCL (29, 6) · SDA (29, 7)	to board B, J7

Board A: hole coordinates wiring list

#	Kind	Net	From → to	Holes / length
1	bare	**B1**	(1, 4) → (2, 4)	1
2	bare	**B1**	(3, 3) → (4, 3)	1
3	bare	**N1**	(8, 3) → (9, 3)	1
4	bare	**N1**	(9, 3) → (9, 4)	1
5	bare	**A0**	(13, 2) → (14, 2) via (13, 3), (14, 3)	3
6	bare	**GND**	(13, 4) → (15, 4)	2
7	bare	**A0**	(19, 10) → (20, 10)	1
8	bare	**B2**	(1, 5) → (2, 5)	1
9	bare	**B2**	(3, 8) → (4, 8)	1
10	bare	**N2**	(8, 8) → (9, 8)	1
11	bare	**N2**	(9, 8) → (9, 9)	1

#	Kind	Net	From → to	Holes / length
12	bare	A1	(13, 7) → (14, 7) via (13, 8), (14, 8)	3
13	bare	GND	(13, 9) → (15, 9)	2
14	bare	A1	(19, 11) → (20, 11)	1
15	bare	B3	(1, 6) → (2, 6)	1
16	bare	B3	(3, 13) → (4, 13)	1
17	bare	N3	(8, 13) → (9, 13)	1
18	bare	N3	(9, 13) → (9, 14)	1
19	bare	A2	(13, 12) → (14, 12) via (13, 13), (14, 13)	3
20	bare	GND	(13, 14) → (15, 14)	2
21	bare	A2	(19, 12) → (20, 12)	1
22	bare	B+	(1, 7) → (2, 7)	1
23	bare	B+	(3, 18) → (4, 18)	1
24	bare	N4	(8, 18) → (9, 18)	1
25	bare	N4	(9, 18) → (9, 19)	1
26	bare	A3	(13, 17) → (14, 17) via (13, 18), (14, 18)	3
27	bare	GND	(13, 19) → (15, 19)	2
28	bare	A3	(19, 13) → (20, 13)	1
29	bare	GND	(15, 1) → (15, 21)	20
30	bare	3V3	(17, 1) → (17, 21)	20
31	bare	3V3	(17, 4) → (29, 4)	12
32	bare	GND	(15, 5) → (16, 5)	1
33	bare	GND	(19, 5) → (29, 5)	10
34	bare	SCL	(20, 6) → (29, 6)	9
35	bare	SDA	(20, 7) → (29, 7)	9
36	bare	GND	(20, 8) → (21, 8)	1
37	insulated	B1	(2, 4) → (3, 3)	cut 19 mm
38	insulated	A0	(13, 2) → (19, 10)	cut 40 mm
39	insulated	B2	(2, 5) → (3, 8)	cut 23 mm
40	insulated	A1	(13, 7) → (19, 11)	cut 33 mm
41	insulated	B3	(2, 6) → (3, 13)	cut 33 mm
42	insulated	A2	(13, 12) → (19, 12)	cut 30 mm
43	insulated	B+	(2, 7) → (3, 18)	cut 43 mm
44	insulated	A3	(13, 17) → (19, 13)	cut 33 mm
45	insulated	GND	(16, 5) → (19, 5)	cut 23 mm
46	insulated	GND	(21, 8) → (22, 5)	cut 23 mm

9. Board B: ESP32, temperatures, current, relays and alarm

The ESP32 sits in two 15-pin female headers with its USB end over the left edge, so the console cable fits with the board in the case. The five NTC dividers run down the right side with their connectors on the edge; the INA226 module, the relay and buzzer connectors and the 5 V input sit along the bottom edge.

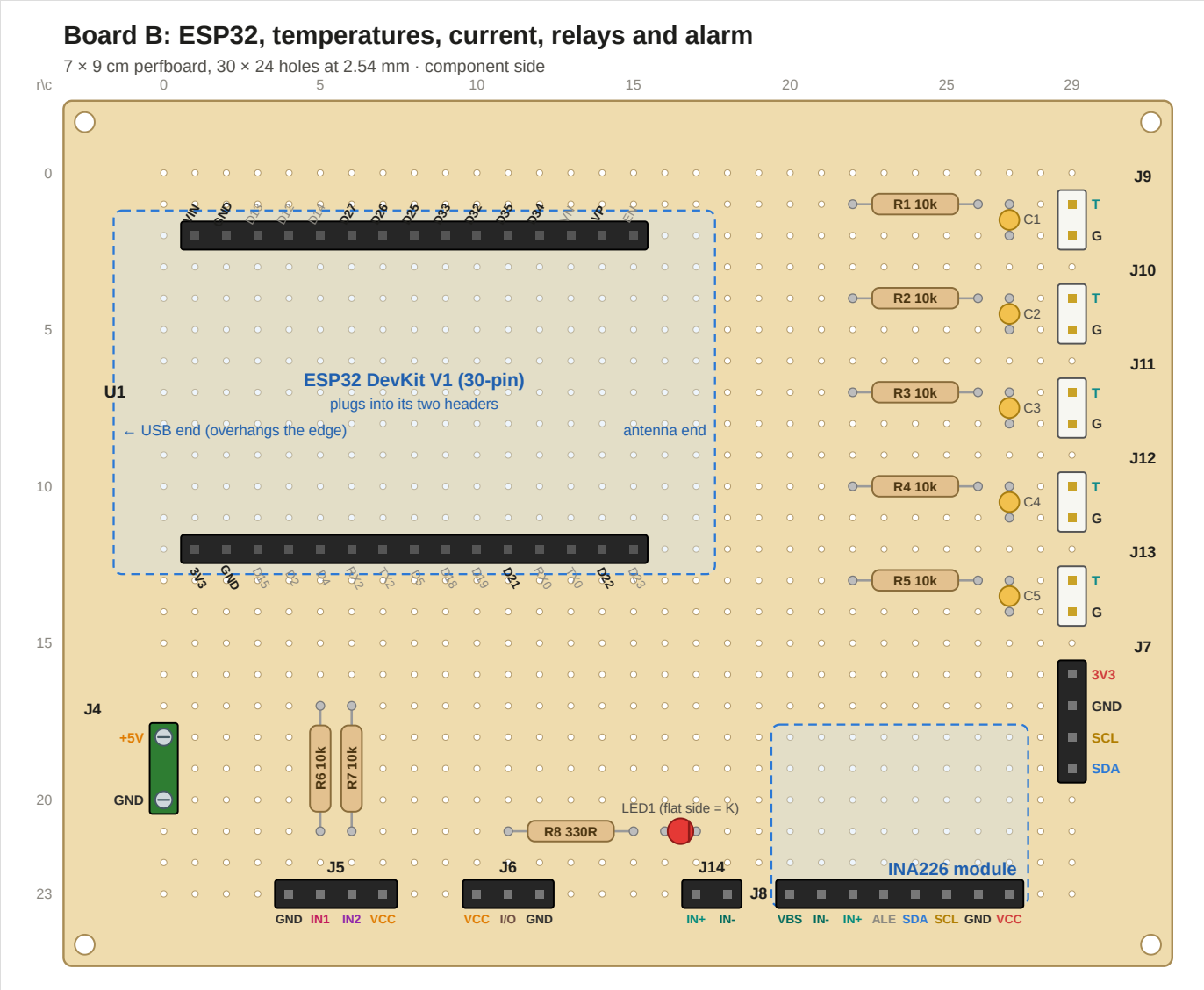


Figure 9. Board B, component side.

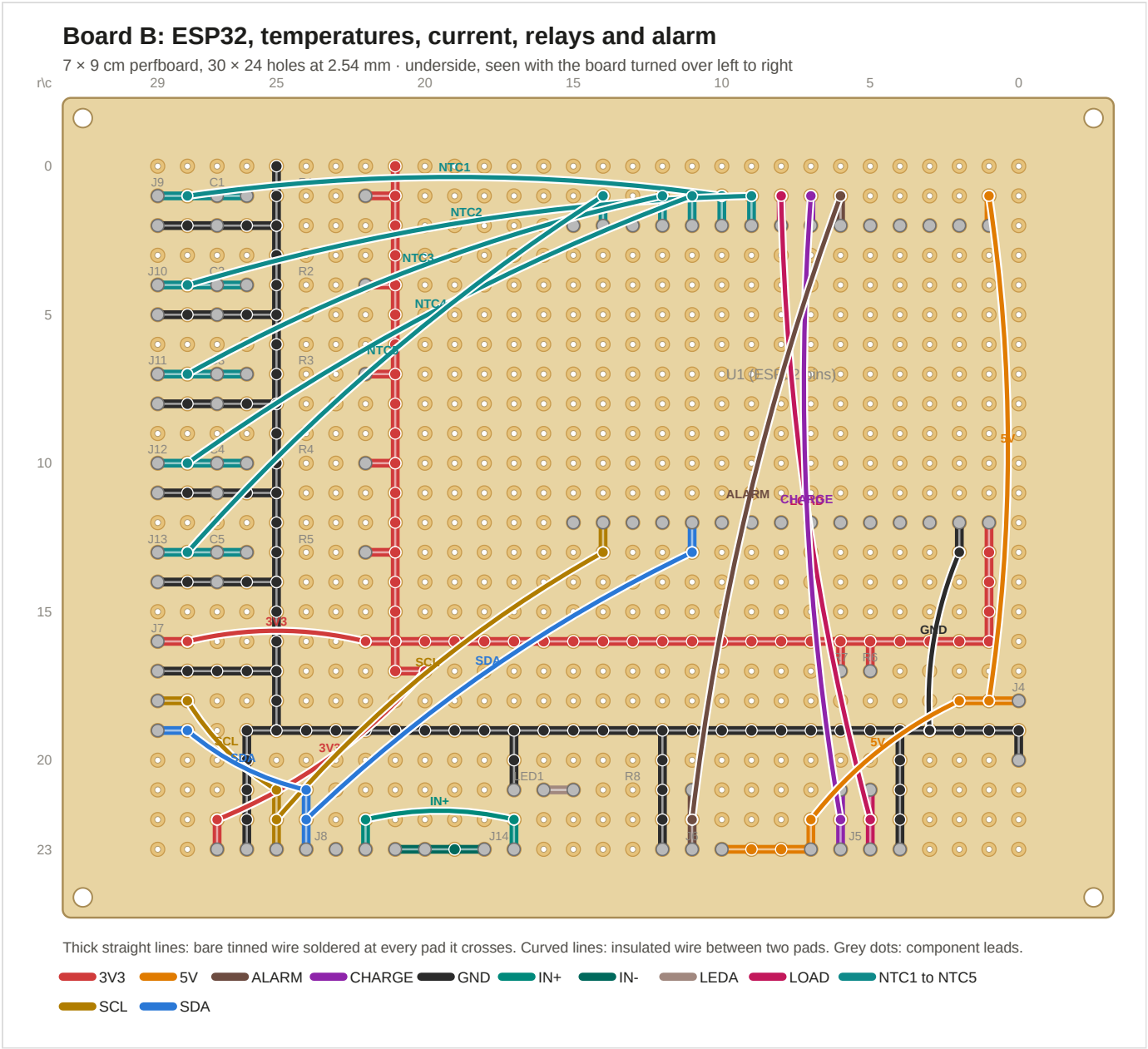


Figure 10. Board B, underside (turned over left to right).

Board B: Pin-to-pin wiring list (schematic connections)

If you placed parts at different hole coordinates, wire the pins listed in each row together:

Net	Pins to join together	Function
3V3	**ESP32** 3V3 · **R1** pin 1 (10k) · **R2** pin 1 (10k) · **R3** pin 1 (10k) · **R4** pin 1 (10k) · **R5** pin 1 (10k) · **INA226** VCC · **Board link J7** pin 3V3 · **R6** pin 1 (10k) · **R7** pin 1 (10k)	3.3 V logic & sensor supply bus
5V	**ESP32** VIN · **5V terminal J4** +5V · **Relay header J5** VCC · **Buzzer header J6** VCC	5.00 V supply from LM2596 (powers ESP32 VIN, relays, buzzer)
ALARM	**ESP32** D27 · **Buzzer header J6** I/O · **R8** pin 1 (330R)	Alarm output from ESP32 GPIO27 to buzzer and LED resistor
CHARGE	**ESP32** D26 · **Relay header J5** IN2 · **R7** pin 2 (10k)	K_CHARGE relay control (active LOW, pulled up to 3V3)
GND	**ESP32** GND · **ESP32** GND · **C1** pin 2 (100n) · **J9** [NTC1 (cell 1)] pin G · **C2** pin 2 (100n) · **J10** [NTC2 (cell 2)] pin G · **C3** pin 2 (100n) · **J11** [NTC3 (cell 3)] pin G · **C4** pin 2 (100n) · **J12** [NTC4 (cell 4)] pin G · **C5** pin 2 (100n) · **J13** [NTC5 (air)] pin G · **INA226** GND · **Board link J7** pin GND · **5V terminal J4** GND · **Relay header J5** GND · **Buzzer header J6** GND · **LED1** pin K (red LED)	Electronics ground bus (P−)
IN+	**INA226** IN+ · **Shunt Kelvin J14** pin IN+	Current shunt positive Kelvin lead (from fuse side)
IN-	**INA226** VBS · **INA226** IN- · **Shunt Kelvin J14** pin IN-	Current shunt negative Kelvin lead & INA226 bus sense (to load side)
LEDA	**R8** pin 2 (330R) · **LED1** pin A (red LED)	Alarm current feed through 330 Ω to LED1 anode
LOAD	**ESP32** D25 · **Relay header J5** IN1 · **R6** pin 2 (10k)	K_LOAD relay control (active LOW, pulled up to 3V3)
NTC1	**ESP32** D32 · **R1** pin 2 (10k) · **C1** pin 1 (100n) · **J9** [NTC1 (cell 1)] pin T	Cell 1 NTC divider junction to ESP32 GPIO32 (ADC1_CH4)
NTC2	**ESP32** D33 · **R2** pin 2 (10k) · **C2** pin 1 (100n) · **J10** [NTC2 (cell 2)] pin T	Cell 2 NTC divider junction to ESP32 GPIO33 (ADC1_CH5)
NTC3	**ESP32** D34 · **R3** pin 2 (10k) · **C3** pin 1 (100n) · **J11** [NTC3 (cell 3)] pin T	Cell 3 NTC divider junction to ESP32 GPIO34 (ADC1_CH6)
NTC4	**ESP32** D35 · **R4** pin 2 (10k) · **C4** pin 1 (100n) · **J12** [NTC4 (cell 4)] pin T	Cell 4 NTC divider junction to ESP32 GPIO35 (ADC1_CH7)
NTC5	**ESP32** VP · **R5** pin 2 (10k) · **C5** pin 1 (100n) · **J13** [NTC5 (air)] pin T	Ambient NTC divider junction to ESP32 GPIO36 / VP (ADC1_CH0)

Net	Pins to join together	Function
SCL	**ESP32** D22 · **INA226** SCL · **Board link J7** pin SCL	I2C clock line (ESP32 GPIO22 to ADS1115 & INA226)
SDA	**ESP32** D21 · **INA226** SDA · **Board link J7** pin SDA	I2C data line (ESP32 GPIO21 to ADS1115 & INA226)

> **Temporary ACS712 wiring:** If the INA226 module is not available, leave the 1 × 8 female header J8 and J14 empty. Connect the ACS712 module's `OUT` to ESP32 pin **VN (GPIO39)**, `VCC` to 3.3 V, `GND` to GND, and place terminals `IP+` and `IP−` in series with the pack positive after fuse F1. See section 9.1 below.

Board B: placement

Ref	Part	Pins and holes (c, r)	Note
U1	ESP32 DevKit V1 (30-pin)	top row r = 2, bottom row r = 12, columns 1–15; used pins: VIN (1, 2), GND (2, 2), D27 (6, 2), D26 (7, 2), D25 (8, 2), D33 (9, 2), D32 (10, 2), D35 (11, 2), D34 (12, 2), VP (14, 2), 3V3 (1, 12), GND (2, 12), D21 (11, 12), D22 (14, 12)	2 x 15 female headers, rows 10 holes apart
R1	10k	1 (22, 1) · 2 (26, 1)	
C1	100n	1 (27, 1) · 2 (27, 2)	
J9	JST-XH 2-pin	T (29, 1) · G (29, 2)	NTC1 (cell 1)
R2	10k	1 (22, 4) · 2 (26, 4)	
C2	100n	1 (27, 4) · 2 (27, 5)	
J10	JST-XH 2-pin	T (29, 4) · G (29, 5)	NTC2 (cell 2)
R3	10k	1 (22, 7) · 2 (26, 7)	
C3	100n	1 (27, 7) · 2 (27, 8)	
J11	JST-XH 2-pin	T (29, 7) · G (29, 8)	NTC3 (cell 3)
R4	10k	1 (22, 10) · 2 (26, 10)	
C4	100n	1 (27, 10) · 2 (27, 11)	
J12	JST-XH 2-pin	T (29, 10) · G (29, 11)	NTC4 (cell 4)
R5	10k	1 (22, 13) · 2 (26, 13)	
C5	100n	1 (27, 13) · 2 (27, 14)	
J13	JST-XH 2-pin	T (29, 13) · G (29, 14)	NTC5 (air)
J8	INA226 module	VBS (20, 23) · IN- (21, 23) · IN+ (22, 23) · ALE (23, 23) · SDA (24, 23) · SCL (25, 23) · GND (26, 23) · VCC (27, 23)	1 x 8 female header; module stands over the board
J14	2-pin header	IN+ (17, 23) · IN- (18, 23)	shunt sense pair
J7	4-pin header	3V3 (29, 16) · GND (29, 17) · SCL (29, 18) · SDA (29, 19)	to board A, J3
J4	2-pin screw terminal	+5V (0, 18) · GND (0, 20)	from the LM2596 (5.00 V)
J5	4-pin header	GND (4, 23) · IN1 (5, 23) · IN2 (6, 23) · VCC (7, 23)	to the relay module
R6	10k	1 (5, 17) · 2 (5, 21)	
R7	10k	1 (6, 17) · 2 (6, 21)	
J6	3-pin header	VCC (10, 23) · I/O (11, 23) · GND (12, 23)	to the buzzer module

Ref	Part	Pins and holes (c, r)	Note
R8	330R	1 (11, 21) · 2 (15, 21)	sized for the LED
LED1	red LED	A (16, 21) · K (17, 21)	

Board B: hole coordinates wiring list

#	Kind	Net	From → to	Holes / length
1	bare	**5V**	(1, 2) → (1, 1)	1
2	bare	**ALARM**	(6, 2) → (6, 1)	1
3	bare	**CHARGE**	(7, 2) → (7, 1)	1
4	bare	**LOAD**	(8, 2) → (8, 1)	1
5	bare	**NTC2**	(9, 2) → (9, 1)	1
6	bare	**NTC1**	(10, 2) → (10, 1)	1
7	bare	**NTC4**	(11, 2) → (11, 1)	1
8	bare	**NTC3**	(12, 2) → (12, 1)	1
9	bare	**NTC5**	(14, 2) → (14, 1)	1
10	bare	**3V3**	(1, 12) → (1, 16)	4
11	bare	**3V3**	(1, 16) → (21, 16)	20
12	bare	**GND**	(2, 12) → (2, 13)	1
13	bare	**SDA**	(11, 12) → (11, 13)	1
14	bare	**SCL**	(14, 12) → (14, 13)	1
15	bare	**3V3**	(21, 0) → (21, 17)	17
16	bare	**GND**	(25, 0) → (25, 19)	19
17	bare	**3V3**	(21, 1) → (22, 1)	1
18	bare	**NTC1**	(26, 1) → (29, 1)	3
19	bare	**GND**	(25, 2) → (29, 2)	4
20	bare	**3V3**	(21, 4) → (22, 4)	1
21	bare	**NTC2**	(26, 4) → (29, 4)	3
22	bare	**GND**	(25, 5) → (29, 5)	4
23	bare	**3V3**	(21, 7) → (22, 7)	1
24	bare	**NTC3**	(26, 7) → (29, 7)	3
25	bare	**GND**	(25, 8) → (29, 8)	4
26	bare	**3V3**	(21, 10) → (22, 10)	1
27	bare	**NTC4**	(26, 10) → (29, 10)	3
28	bare	**GND**	(25, 11) → (29, 11)	4
29	bare	**3V3**	(21, 13) → (22, 13)	1
30	bare	**NTC5**	(26, 13) → (29, 13)	3
31	bare	**GND**	(25, 14) → (29, 14)	4
32	bare	**IN-**	(18, 23) → (21, 23)	3
33	bare	**IN+**	(17, 23) → (17, 22)	1

#	Kind	Net	From → to	Holes / length
34	bare	IN+	(22, 22) → (22, 23)	1
35	bare	GND	(26, 23) → (25, 19) via (26, 22), (26, 21), (26, 20), (26, 19)	5
36	bare	3V3	(27, 23) → (27, 22)	1
37	bare	3V3	(20, 17) → (21, 17)	1
38	bare	SDA	(24, 21) → (24, 23)	2
39	bare	SCL	(25, 21) → (25, 23)	2
40	bare	3V3	(21, 16) → (22, 16)	1
41	bare	3V3	(28, 16) → (29, 16)	1
42	bare	GND	(25, 17) → (29, 17)	4
43	bare	SCL	(28, 18) → (29, 18)	1
44	bare	SDA	(28, 19) → (29, 19)	1
45	bare	5V	(0, 18) → (2, 18)	2
46	bare	GND	(0, 20) → (3, 19) via (0, 19), (1, 19), (2, 19)	4
47	bare	GND	(3, 19) → (25, 19)	22
48	bare	3V3	(5, 16) → (5, 17)	1
49	bare	3V3	(6, 16) → (6, 17)	1
50	bare	LOAD	(5, 21) → (5, 23)	2
51	bare	CHARGE	(6, 21) → (6, 23)	2
52	bare	GND	(4, 19) → (4, 23)	4
53	bare	5V	(7, 22) → (7, 23)	1
54	bare	5V	(7, 23) → (10, 23)	3
55	bare	GND	(12, 23) → (12, 19)	4
56	bare	ALARM	(11, 21) → (11, 23)	2
57	bare	LEDA	(15, 21) → (16, 21)	1
58	bare	GND	(17, 21) → (17, 19)	2
59	insulated	GND	(2, 13) → (3, 19)	cut 30 mm
60	insulated	NTC1	(10, 1) → (28, 1)	cut 61 mm
61	insulated	NTC2	(9, 1) → (28, 4)	cut 64 mm
62	insulated	NTC3	(12, 1) → (28, 7)	cut 58 mm
63	insulated	NTC4	(11, 1) → (28, 10)	cut 64 mm
64	insulated	NTC5	(14, 1) → (28, 13)	cut 62 mm
65	insulated	IN+	(17, 22) → (22, 22)	cut 28 mm
66	insulated	3V3	(27, 22) → (20, 17)	cut 37 mm
67	insulated	SDA	(11, 13) → (24, 22)	cut 55 mm
68	insulated	SCL	(14, 13) → (25, 22)	cut 51 mm
69	insulated	3V3	(22, 16) → (28, 16)	cut 30 mm
70	insulated	SCL	(28, 18) → (25, 21)	cut 26 mm
71	insulated	SDA	(28, 19) → (24, 21)	cut 26 mm

#	Kind	Net	From → to	Holes / length
72	insulated	5V	(1, 18) → (1, 1)	cut 58 mm
73	insulated	LOAD	(8, 1) → (5, 22)	cut 69 mm
74	insulated	CHARGE	(7, 1) → (6, 22)	cut 68 mm
75	insulated	5V	(2, 18) → (7, 22)	cut 31 mm
76	insulated	ALARM	(6, 1) → (11, 22)	cut 70 mm

9.1 Temporary alternative: ACS712 wiring (until INA226 arrives)

If you need to show the prototype before your INA226 module arrives, an ACS712-5A Hall-effect module can be wired as a temporary substitute. The firmware automatically detects if the INA226 is absent and reads the ACS712 on ADC1 channel 3 (GPIO39 / VN). Solder female headers J8 and J14 on Board B as normal; they stay unpopulated until the INA226 is plugged in.

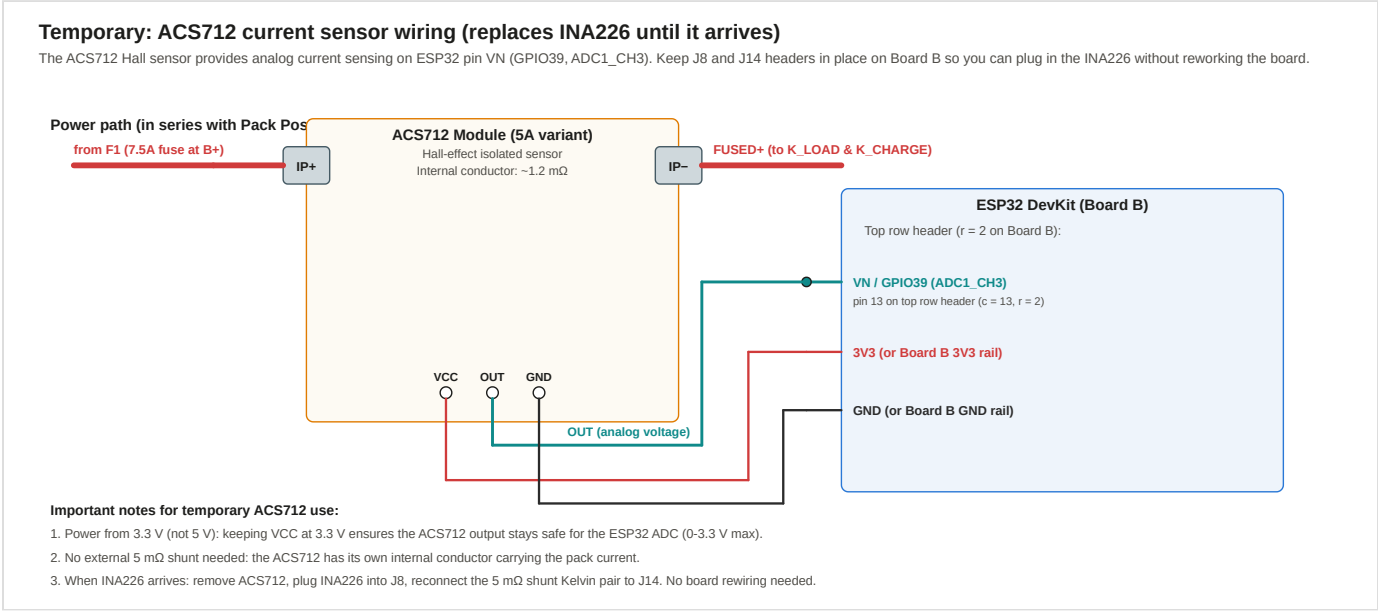


Figure 10a. Temporary ACS712 current sensor wiring to Board B.

ACS712 Terminal	Connects to	Notes
VCC	Board B 3V3 rail (or J8 pin 8)	Power from 3.3 V so analog output never exceeds 3.3 V
GND	Board B GND rail (or J8 pin 7)	Shared electronics ground (P−)
OUT	ESP32 pin VN (GPIO39 · ADC1_CH3)	Top row header pin 13 (c = 13, r = 2)
IP+	Pack positive after Fuse F1	16 AWG power wire from the fuse output
IP−	Feeder bus FUSED+	To K_LOAD and K_CHARGE relay contacts

Reverting to INA226: Disconnect the ACS712 wires, plug the INA226 module into header J8, and connect the 5 mΩ shunt Kelvin pair to J14. The firmware prefers INA226 automatically whenever it is present.

10. Module pin orders

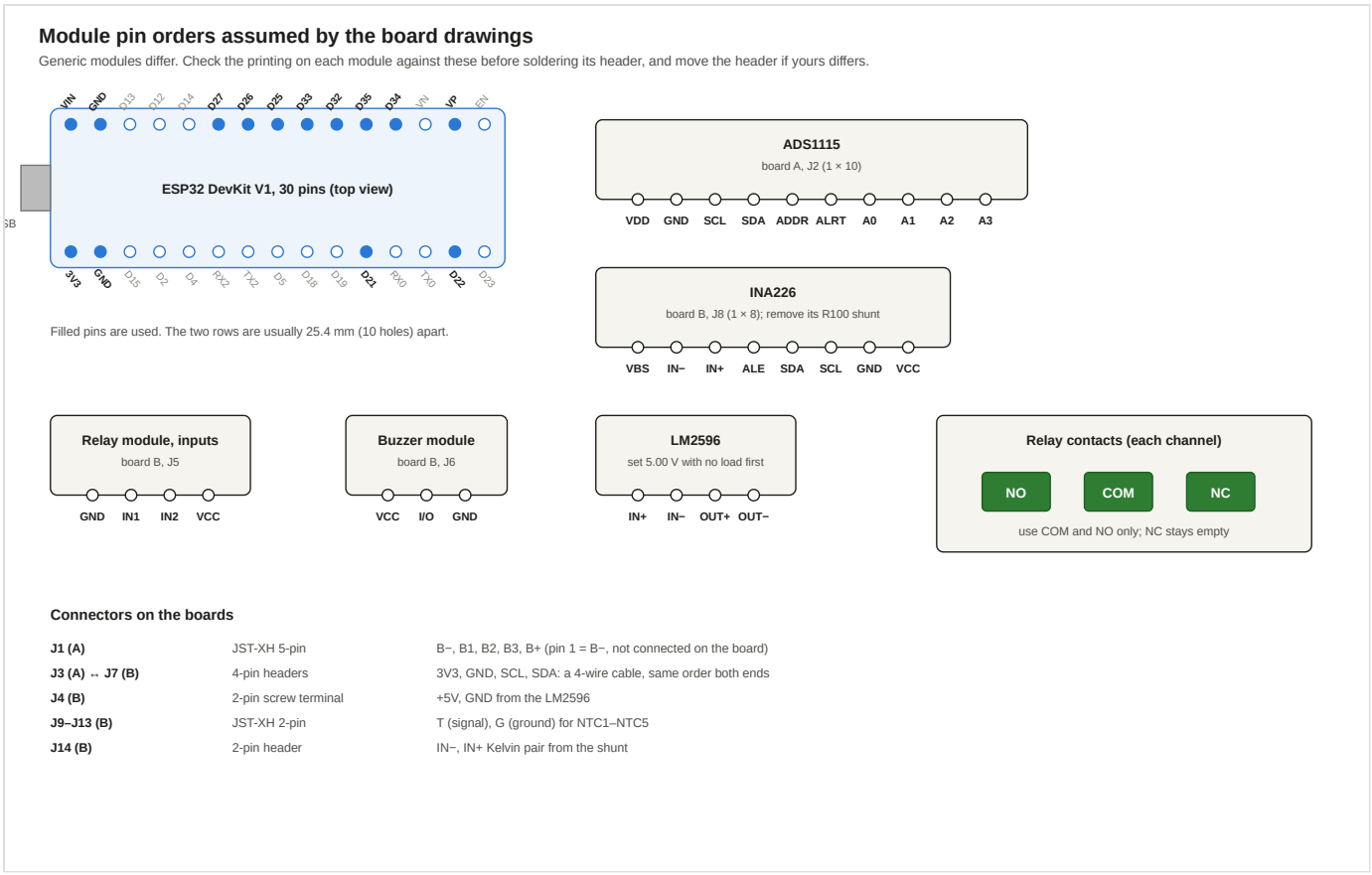


Figure 11. The pin orders the board layouts assume.

11. Enclosure layout

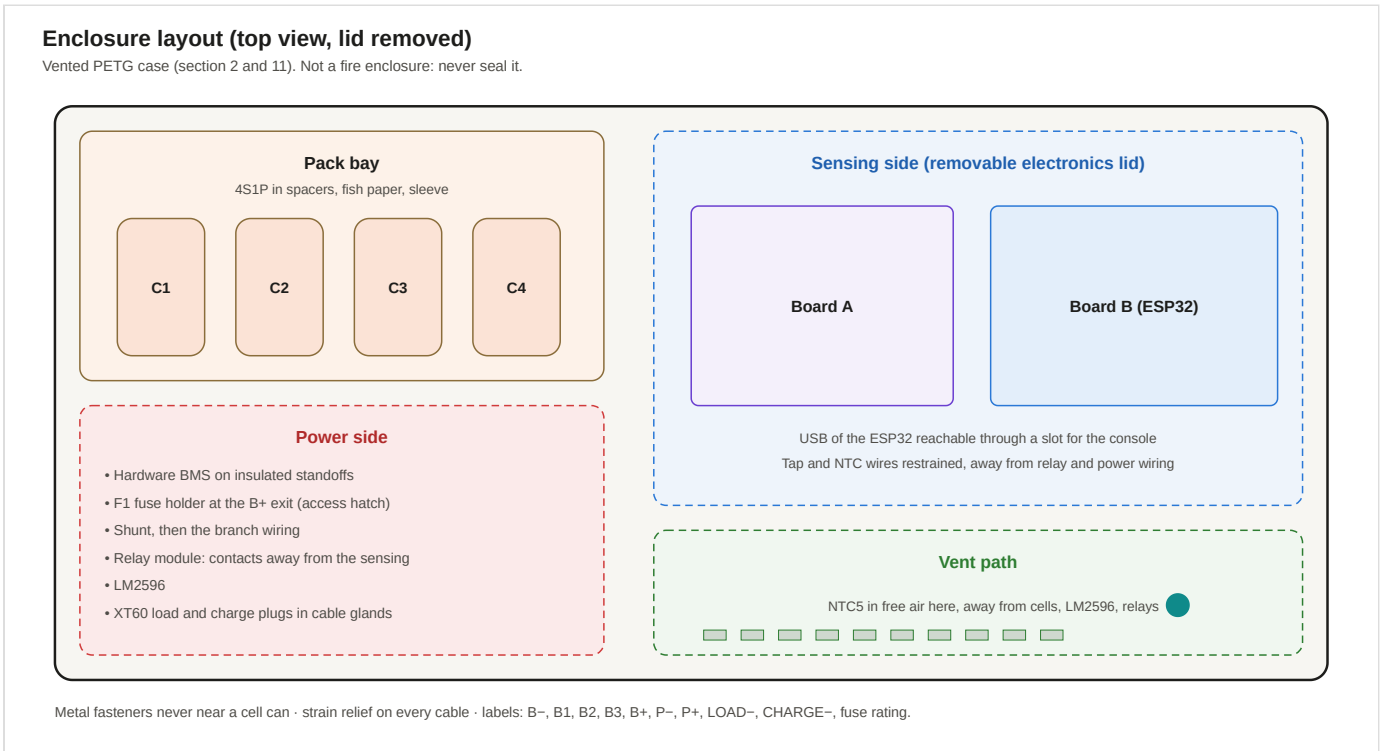


Figure 12. Vented PETG enclosure: pack bay, power side, sensing side, vent path.

12. Assembly, step by step

Board A

1. Mark the top-left corner of the component side with a pen, so the hole coordinates stay the right way round.
2. Fit and solder the low parts first: R1–R12, then C1–C4. Leave D1–D4 empty: the clamp part is chosen after the Gate 1 leakage test.
3. Fit J1, the 1 × 10 female header J2 and the 4-pin header J3.
4. Underside: lay the two long buses (GND column 15, 3V3 column 17) as bare wire, then every bare bridge of the wiring list, then the insulated links. Tick each line of the wiring list as you go.
5. Clean the flux off with isopropyl alcohol: residue leaks current across the 330 kΩ dividers.
6. Do the checks of section 13 before fitting the ADS1115.

Board B

1. Fit R1–R8, C1–C5 and LED1 (flat side, the cathode, towards column 17).
2. Fit the headers: two 1 × 15 female headers for the ESP32 (dry-fit the DevKit first: most have their rows 25.4 mm, 10 holes, apart), the 1 × 8 female header J8, J5, J6, J7, J14, the five 2-pin JST sockets and the screw terminal J4.
3. Underside: the rails (3V3 along row 16 and column 21, GND along row 19 and column 25), then the bridges, then the insulated links.
4. Clean the flux, then do the checks of section 13 before plugging in the ESP32 or the INA226.

Then, gate by gate

1. Gate 1: plug in the ESP32, the ADS1115 and the INA226, join J3 to J7, flash the firmware, calibrate (section 14).
2. Gate 2: set the LM2596 to 5.00 V with nothing connected and load-test it; then wire J4, the relay module (J5) and the buzzer (J6).
3. Gates 3–8: the BMS harness, cell matching, welding and the first power-up, as the safe build sequence says.

13. Checks before power

With a multimeter, with no modules plugged in and the clamps not fitted. A reading far from the value means a wrong part or a missing or extra bridge: find it before going on.

Board A: resistance

Between	Expected	Why
J1 B1 → J3 GND (and B2, B3, B+ on their channels)	386 kΩ	330 kΩ + 56 kΩ
J1 B1 → J1 B2 (and each neighbouring pair)	772 kΩ	no tap is shorted to another
J2 A0 → J3 GND (and A1, A2, A3)	66 kΩ	10 kΩ + 56 kΩ
J1 B1 → J2 A0	340 kΩ	330 kΩ + 10 kΩ
J3 3V3 → J3 GND	open	before the clamps are fitted
J1 B− → anything	open	B− is not connected on the board

Board B: resistance

Between	Expected	Why
J7 3V3 → J7 GND, J4 +5V → J4 GND, J4 +5V → J7 3V3	open	no short between supplies
J9 T → J7 3V3 (and J10–J13 T)	10 kΩ	NTC pull-ups R1–R5
J5 IN1 → J7 3V3, J5 IN2 → J7 3V3	10 kΩ	relay pull-ups R6, R7
J6 I/O → J6 GND, meter on diode test (red on I/O)	about 2 V (1.8–2.4 V)	the LED's forward voltage plus R8 (the LED may glow)
J14 IN+ → J14 IN−	open	the sense pair is not shorted

Continuity: every pin that must be joined

Board A

Net	Pins	Reading
3V3	J2 VDD, J3 3V3	0 Ω (beeps)
GND	J2 GND, J2 ADDR, J3 GND	0 Ω (beeps)
SCL	J2 SCL, J3 SCL	0 Ω (beeps)
SDA	J2 SDA, J3 SDA	0 Ω (beeps)

Board B

Net	Pins	Reading
3V3	U1 3V3, J8 VCC, J7 3V3	0 Ω (beeps)
5V	U1 VIN, J4 +5V, J5 VCC, J6 VCC	0 Ω (beeps)
ALARM	U1 D27, J6 I/O	0 Ω (beeps)
CHARGE	U1 D26, J5 IN2	0 Ω (beeps)
GND	U1 GND, U1 GND, J9 G, J10 G, J11 G, J12 G, J13 G, J8 GND, J7 GND, J4 GND, J5 GND, J6 GND	0 Ω (beeps)
IN+	J8 IN+, J14 IN+	0 Ω (beeps)

Net	Pins	Reading
IN-	J8 VBS, J8 IN-, J14 IN-	0 Ω (beeps)
LOAD	U1 D25, J5 IN1	0 Ω (beeps)
NTC1	U1 D32, J9 T	0 Ω (beeps)
NTC2	U1 D33, J10 T	0 Ω (beeps)
NTC3	U1 D34, J11 T	0 Ω (beeps)
NTC4	U1 D35, J12 T	0 Ω (beeps)
NTC5	U1 VP, J13 T	0 Ω (beeps)
SCL	U1 D22, J8 SCL, J7 SCL	0 Ω (beeps)
SDA	U1 D21, J8 SDA, J7 SDA	0 Ω (beeps)

14. Bring-up, gate by gate

The full procedure is in `docs/bring-up.md`; the pass and stop rules are in the safe build sequence.

Gate	What	Firmware and tools	Pass when
0	Design frozen	Build the firmware once; measure the charger plug	every part has a purpose and rating
1	Sensing, no cells	`raw`, `cal`, tools/calibrate.py	taps ±20 mV; current ±2 % or ±20 mA; every NTC fault detected
2	Relays fail-safe	bench build: `sim on`, `test hang`, `test crash`	both contacts open on power-off, boot, reset, watchdog, I2C loss
3	BMS harness	bench channels; `status`	tap order and polarity; each cutoff opens only its branch
4	Cell matching	`model r0 …`	20 mV, 3 % capacity, 10 % resistance
5–7	Mechanics, weld, pack	none	no conductor can rub a cell; sound welds
8	First power-up	`status`, tools/serial_log.py	stable readings at rest for 30 min
9	Controlled discharge	tools/fit_thermal.py, `model thermal …`	abnormal heat < 0.25 °C/min; 5-min error ≤ 0.5 °C RMS
10	Controlled charge	tools/ocv_table.py, `model ocv …`	no unexpected 4.15 V; CC/CV taper
11	Predictive layer	`bms_sim run all`; dummy heater; 47 Ω soft short	warnings and trips as the architecture says
12	Enclosure and release	normal build, model fitted	every gate signed; guide accepts

Firmware and tools

- Build and flash: `idf.py build`, `idf.py -p PORT flash monitor` (ESP-IDF v5.5).
- Serial console at 115200 baud: `status`, `raw`, `cal …`, `model …`, `log 1`, `reset`, `test hang`, `test crash`, `wifi off`.
- Dashboard over Wi-Fi: the ESP32's own access point "PredictiveBMS", http://192.168.4.1/. Every protection works with Wi-Fi off.

15. The simulation

Before any cell is connected, the supervisor is tested against a simulated 4S pack: the same code that runs on the ESP32, with cells, heat, a hardware BMS, sensors with noise, and every fault Gate 11 asks for. A second copy of the pack runs alongside with only fixed limits, as a conventional BMS has, to show what the predictive layer adds. Start it with `python3 tools/demo.py`; it opens the dashboard in the browser.

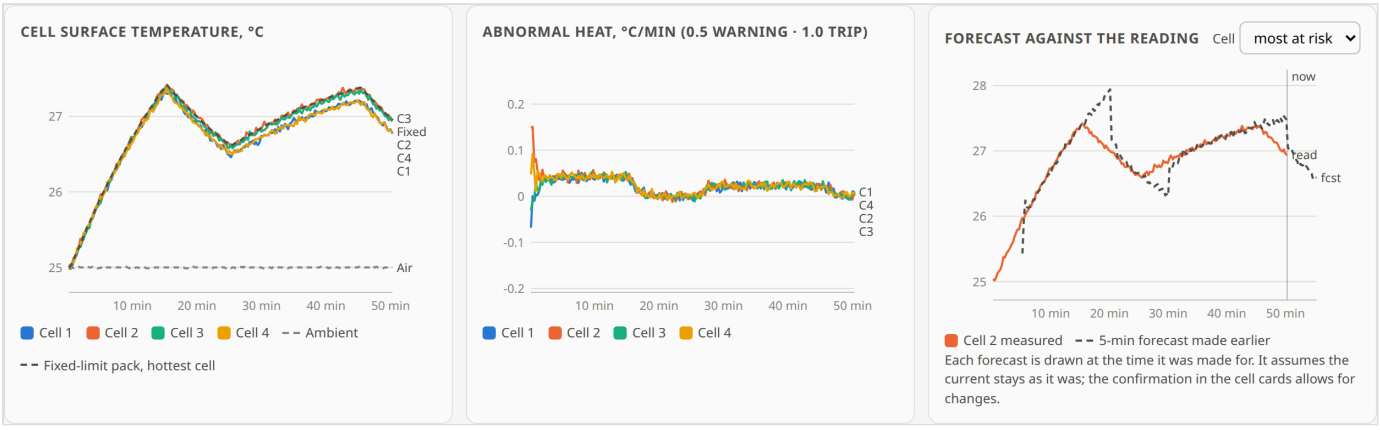


Figure 13. Normal use: cell temperatures, abnormal heat near zero, and each 5-minute forecast drawn against the reading it predicted.

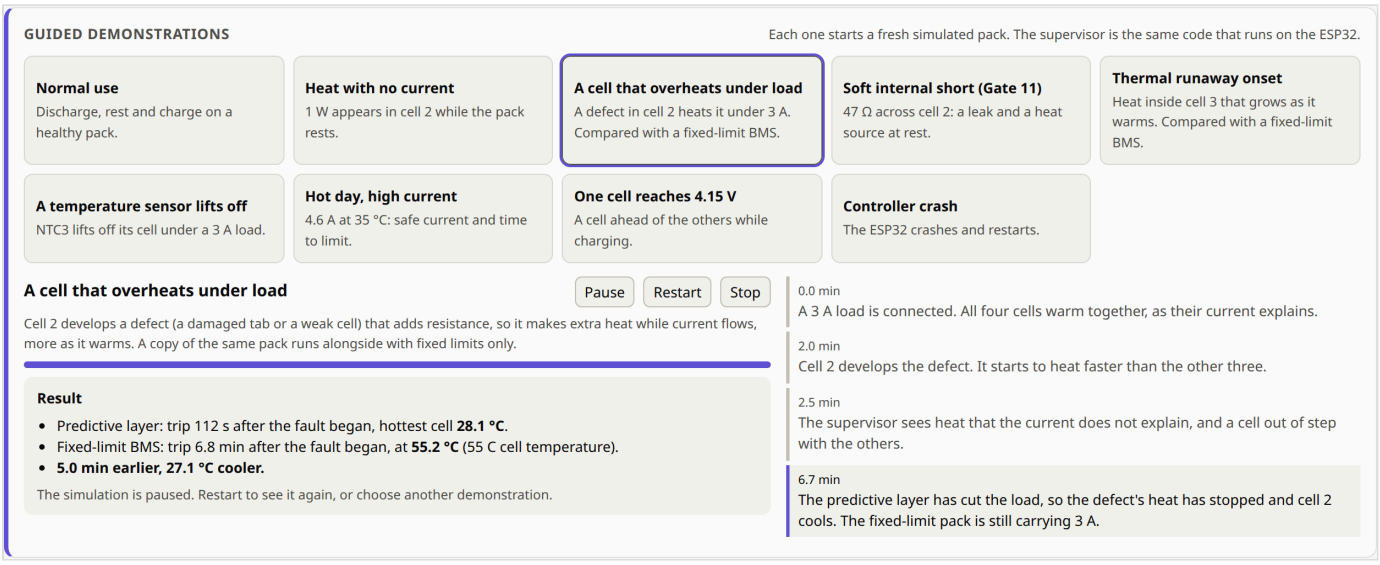


Figure 14. A guided demonstration: a cell defect under a 3 A load. The narration follows each step; the result is measured by the simulator.

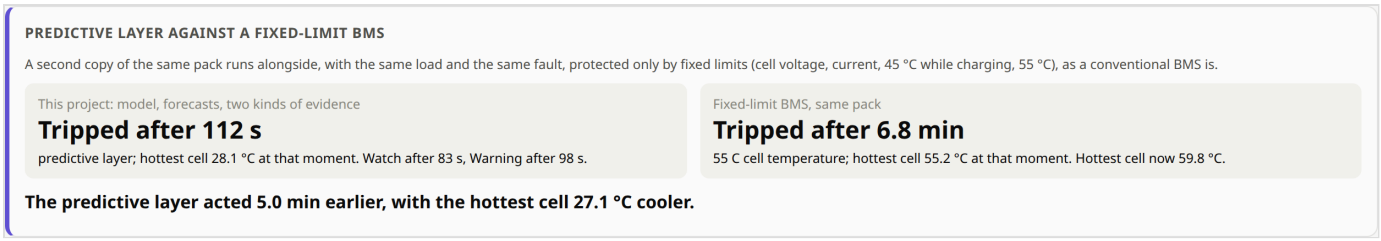


Figure 15. The same run against a fixed-limit BMS on the same pack: the predictive layer acts minutes earlier, with the cell far cooler.

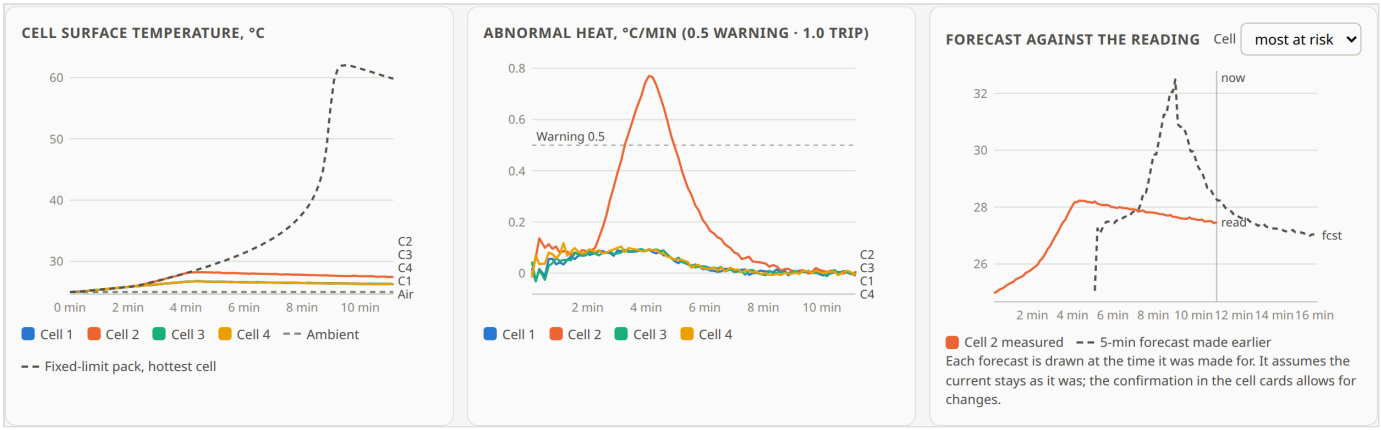


Figure 16. Charts of that run. The defective cell (cell 2) is cut off early and cools; the fixed-limit pack's hottest cell (dashed) climbs to 55 °C.

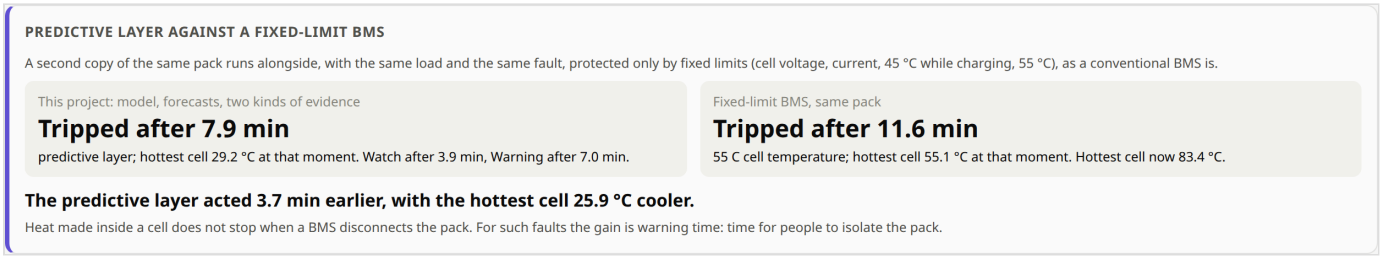


Figure 17. Thermal runaway onset (heat inside a cell that grows as it warms): the warning comes minutes before a 55 °C limit. Heat made inside a cell does not stop when the pack is disconnected; the gain is warning time.

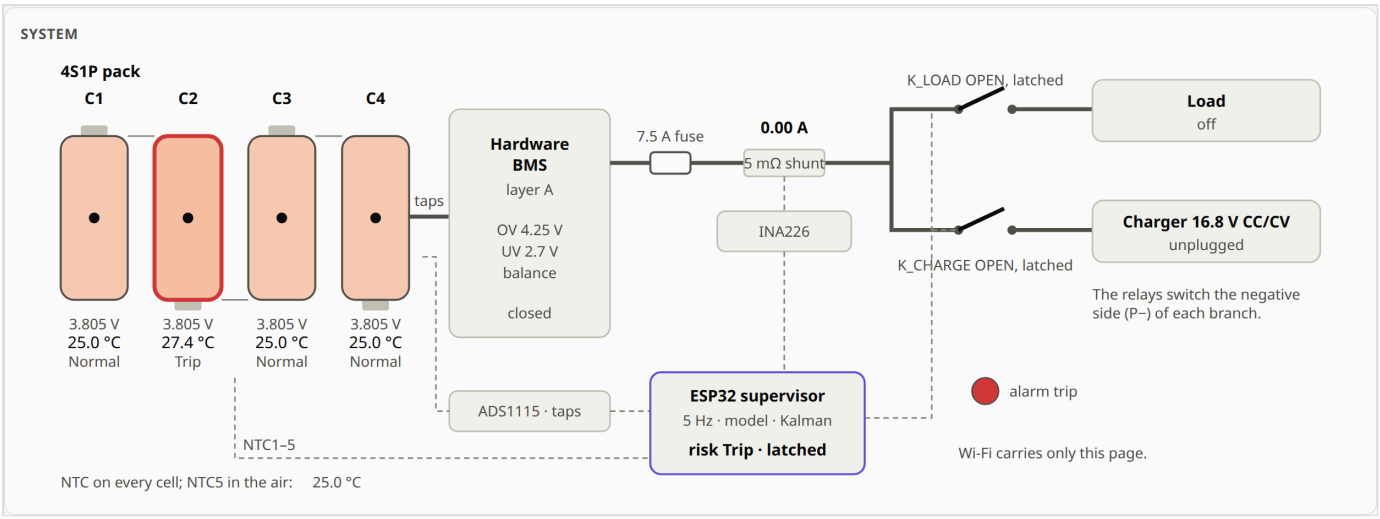


Figure 18. Live system diagram after a trip: cell colours follow temperature; both relays open and latched; alarm on.

Cell 1	Normal
Voltage	3.7022 V
Resistance (online, model)	– · 34.9 mΩ (0 steps)
Surface, measured	25.01 °C
Core, estimated	25.00 °C
Rate of rise	0.00 °C/min
Abnormal heat	0.000 W · 0.00 °C/min
Against the other cells	-0.00 °C/min
Forecast 1 min · 5 min	24.99 · 24.98 °C
Last check 1 min · 5 min	-0.03 · -0.03 °C ✓ confirmed
5-min forecast error (hour · all)	0.02 · 0.02 °C (547)
Settles at · time to limit	25.0 °C · never at this load
Safe current	5.54 A
Leak at rest · cross-check	none (-20.6 mA)

Cell 2	Watch
Voltage	3.6867 V
Resistance (online, model)	– · 35.0 mΩ (0 steps)
Surface, measured	30.03 °C
Core, estimated	30.32 °C
Rate of rise	-0.21 °C/min
Abnormal heat	0.000 W · 0.00 °C/min
Against the other cells	-0.00 °C/min
Forecast 1 min · 5 min	29.83 · 29.11 °C
Last check 1 min · 5 min	-0.01 · -0.16 °C ✓ confirmed
5-min forecast error (hour · all)	1.25 · 1.04 °C (547)
Settles at · time to limit	25.0 °C · never at this load
Safe current	5.26 A
Leak at rest · cross-check	62.5 mA · disagree
self-discharge at rest	

Cell 3	Normal
Voltage	3.7023 V
Resistance (online, model)	– · 34.4 mΩ (0 steps)
Surface, measured	25.01 °C
Core, estimated	25.01 °C
Rate of rise	0.02 °C/min
Abnormal heat	0.002 W · 0.00 °C/min
Against the other cells	0.00 °C/min
Forecast 1 min · 5 min	25.01 · 25.02 °C
Last check 1 min · 5 min	0.00 · 0.00 °C ✓ confirmed
5-min forecast error (hour · all)	0.02 · 0.02 °C (547)
Settles at · time to limit	25.1 °C · never at this load
Safe current	5.30 A
Leak at rest · cross-check	none (-20.5 mA)

Cell 4	Normal
Voltage	3.7027 V
Resistance (online, model)	– · 34.9 mΩ (0 steps)
Surface, measured	25.02 °C
Core, estimated	25.01 °C
Rate of rise	0.03 °C/min
Abnormal heat	0.006 W · 0.01 °C/min
Against the other cells	0.01 °C/min
Forecast 1 min · 5 min	25.01 · 25.03 °C
Last check 1 min · 5 min	0.00 · -0.01 °C ✓ confirmed
5-min forecast error (hour · all)	0.02 · 0.02 °C (547)
Settles at · time to limit	25.2 °C · never at this load
Safe current	5.53 A
Leak at rest · cross-check	none (-21.1 mA)

Figure 19. Cell cards at the end of the Gate 11 soft-short demonstration: the leak estimate, the energy cross-check and the confirmation on cell 2.



VERIFICATION: EVERY SCENARIO, EVERY CHECK

30 of 30 pass · 1.4 s | Run again

Each scenario rehearses a test of the build sequence on the simulated pack, with the gate's own pass criteria as checks. Select a row to see its checks.

Result	Scenario	Gate	What it shows
PASS	rest	9	30 minutes at rest: Normal, small abnormal heat, accurate forecasts
PASS	discharge_steps	9	0.25-2 A steps: no flags, forecasts within 0.5 C RMS, resistance within 15 %
PASS	discharge_5a	9, 11	4.8 A for 20 minutes: the heating expected near 5 A is not flagged
PASS	charge	10	1.5 A charge to 4.10 V per cell: no flags, small spread
PASS	heat_no_current	11	1 W in a cell with no current: Warning, then Trip below 40 C
	✓ first trip is predictive_risk (got predictive_risk) ✓ relays: load open, charge open (expected load open, charge open) ✓ the trip names cell 2 ✓ Warning came before the trip ✓ trip 70 s after the heat began ✓ cell 2 was at 26.3 C, far below every fixed temperature limit ✓ alarm on, latched		
PASS	heat_warning	11	0.41 W in a cell: Warning and alarm, both branches stay on
PASS	small_fault	11	0.2 W in a cell: 5-minute confirmation +1 C raises Watch
PASS	cell_outlier	11	0.55 W in one cell under 2 A: trips on the cell-to-cell difference
PASS	self_acceleration	11	heat growing with temperature in every cell: self-acceleration trip
PASS	load_defect	11	a cell defect heating under 3 A: trips below 35 C, then the cell cools
PASS	soft_short	11	47 Ohm across cell 2: leak, heat and confirmation agree; Warning
PASS	internal_leak	11	25 mA internal leak: found electrically, Watch only
PASS	forecast_hot_day	9	4.6 A at 35 C: safe-current and time-to-limit Watch, then load trip
PASS	ntc_lift_off	11	NTC3 lifts under 3 A: sensor-contact fault, both branches open
PASS	ntc_open	1, 11	open NTC: sensor fault after 2 s
PASS	ntc_short	1, 11	shorted NTC: sensor fault after 2 s
PASS	ambient_open	1, 11	open ambient NTC: sensor fault after 2 s
PASS	ntc_stuck	11	flat-line NTC: sensor fault after 60 s
PASS	ads_loss	2	ADS1115 bus loss: both branches open after 2 s
PASS	ina_loss	2	INA226 bus loss: both branches open after 2 s
PASS	spikes	11	single-sample spikes: rejected, no trip
PASS	load_steps	9	0-3 A steps every 20 s: no false trips, resistance tracked
PASS	cell_high_charge	10	one cell reaches 4.15 V on charge: only the charge branch opens
PASS	cell_low_discharge	9	one cell reaches 3.10 V on discharge: only the load branch opens
PASS	overcurrent	9, 10	5.3 A load and 1.8 A charge: each opens only its branch
PASS	charge_overtemp	10	45 C while charging: only the charge branch opens
PASS	temp_spread	11	a 10 C spread (simulated signal, model off): both branches open
PASS	backstop_unfitted	11	model off: the raw 1.0 C/min backstop trips
PASS	reset_rules	2	latches hold through restarts; reset only when the rules allow
PASS	fit_log	9	1 A and 2 A discharges with cooling curves, model off: data for the fit

Figure 20. Verification: every scenario with its checks, run from the dashboard.

Results of the guided demonstrations

Demonstration	What the simulator showed
Normal use	Highest risk level: Normal. Largest abnormal heat: 0.06 °C/min (the Gate 9 limit for healthy use is 0.25). 5-minute forecast error: 0.11, 0.12, 0.12, 0.11 °C RMS for cells 1–4 (Gate 9 asks for 0.5 or less).
Heat with no current	Trip: predictive layer, 58 s after the heat began, with cell 2 at 26.0 °C. Both relays are open and latched; the alarm is on. A fixed-limit BMS: no reaction; hottest cell 27.4 °C, far below 55 °C.
A cell that overheats under load	Predictive layer: trip 112 s after the fault began, hottest cell 28.1 °C. Fixed-limit BMS: trip 6.8 min after the fault began, at 55.2 °C (55 C cell temperature). 5.0 min earlier, 27.1 °C cooler.
Soft internal short (Gate 11)	Leak estimate, cell 2: 73.4 mA (the resistor drew about 79 mA); the other cells showed none. Energy cross-check while the resistor was on: agree. Level reached: Warning, from thermal and electrical evidence agreeing. No trip: a soft fault raises the alarm and keeps both branches on. A fixed-limit BMS: no reaction; the cell warmed only a few degrees.
Thermal runaway onset	Predictive layer: trip 7.9 min after the fault began, hottest cell 29.2 °C. Fixed-limit BMS: trip 11.6 min after the fault began, at 55.1 °C (55 C cell temperature). 3.7 min earlier, 25.9 °C cooler. Disconnecting does not stop heat made inside a cell; the alarm gives people time to isolate the pack.
A temperature sensor lifts off	Fault: NTC lifted off its cell, 6 s after the sensor lifted. Both branches opened: with a sensor it cannot trust, the supervisor does not guess. The other cells were not blamed.
Hot day, high current	The load stopped at 55 °C cell temperature; only the load branch opened, so charging stays possible. Safe continuous current at the end: 4.03 A.
One cell reaches 4.15 V	Trip: cell above 4.15 V. Charge relay open, load relay closed: each limit opens only its own branch.
Controller crash	The events show the latch, the refused reset and the accepted reset. Relays now: load closed, charge closed.

The verification scenarios, all 30 of them, are listed in `docs/simulator.md`; each one rehearses a gate test with the gate's own pass criteria.

16. Circuit and model simulation in KiCad, GNU Octave and Scilab Xcos

Four tools check different parts of the design, and each one is free and open source:

Tool	What it checks	Where
KiCad 10 with its simulator, **ngspice**	The analog circuits: the cell-tap dividers and filters, the NTC dividers, the current shunt, and the cell's thermal model drawn as an electrical circuit	`kicad/`
GNU Octave (the same files run in MATLAB)	The predictive model: the thermal model and the Kalman observer, written a second time from the firmware and compared with it sample by sample	`matlab/`
Scilab Xcos (the free tool like Simulink)	The thermal model and the predictive layer as block diagrams: a fault that a fixed limit misses for half an hour	`scilab/`
The project's simulator (section 15)	The complete firmware: every limit, fault and gate scenario, on a simulated pack	`host/`

The KiCad sheets open in KiCad 10: open the project, then **Inspect → Simulator** and run the analysis written on the sheet. `python3 kicad/run_sims.py` runs all of them from the command line and checks the results; the Octave scripts run with `octave --no-gui --eval "crosscheck"` in `matlab/`; the Xcos models open in Scilab with a double-click and run with the Start button. KiCad's electrical rules check finds no violation on any sheet. All results in this section are simulation results with the part values on the sheets; the gates measure the real circuits.

Simulation checks

Circuit	Check	Result	Limit or unit	
cell taps	highest ADC input at 4 x 4.2 V	2.431	< 3.0 V (Gate 1 stop)	pass
cell taps	tap error per 10 nA of leakage	4.11	mV (for the TBD clamp)	pass
cell taps	settling to 16 bits after a step	64.4	< 200 ms (one 5 Hz sample)	pass
NTC	divider at 25 °C	1.65	1.65 V (half of 3.3 V)	pass
NTC	ADC voltage at the charge trip (45 °C)	1	0.15-2.45 V (ESP32 ADC1, 12 dB)	pass
NTC	ADC voltage at the discharge trip (55 °C)	0.7573	0.15-2.45 V (ESP32 ADC1, 12 dB)	pass
current	shunt voltage at 5 A	25	within +/-81.92 mV (INA226)	pass
current	sign: discharge reads positive	25	> 0	pass
current	error if the module's R100 is left in place	-4.76	% (must be removed; Gate 1 allows 2 %)	pass
thermal	core rise after 1 h of 1 W	29.459	approaches 32 °C (R_core + R_air)	pass
thermal	ngspice against Octave, largest difference	0.0002	< 0.01 °C	pass

16.1 Cell taps (board A)

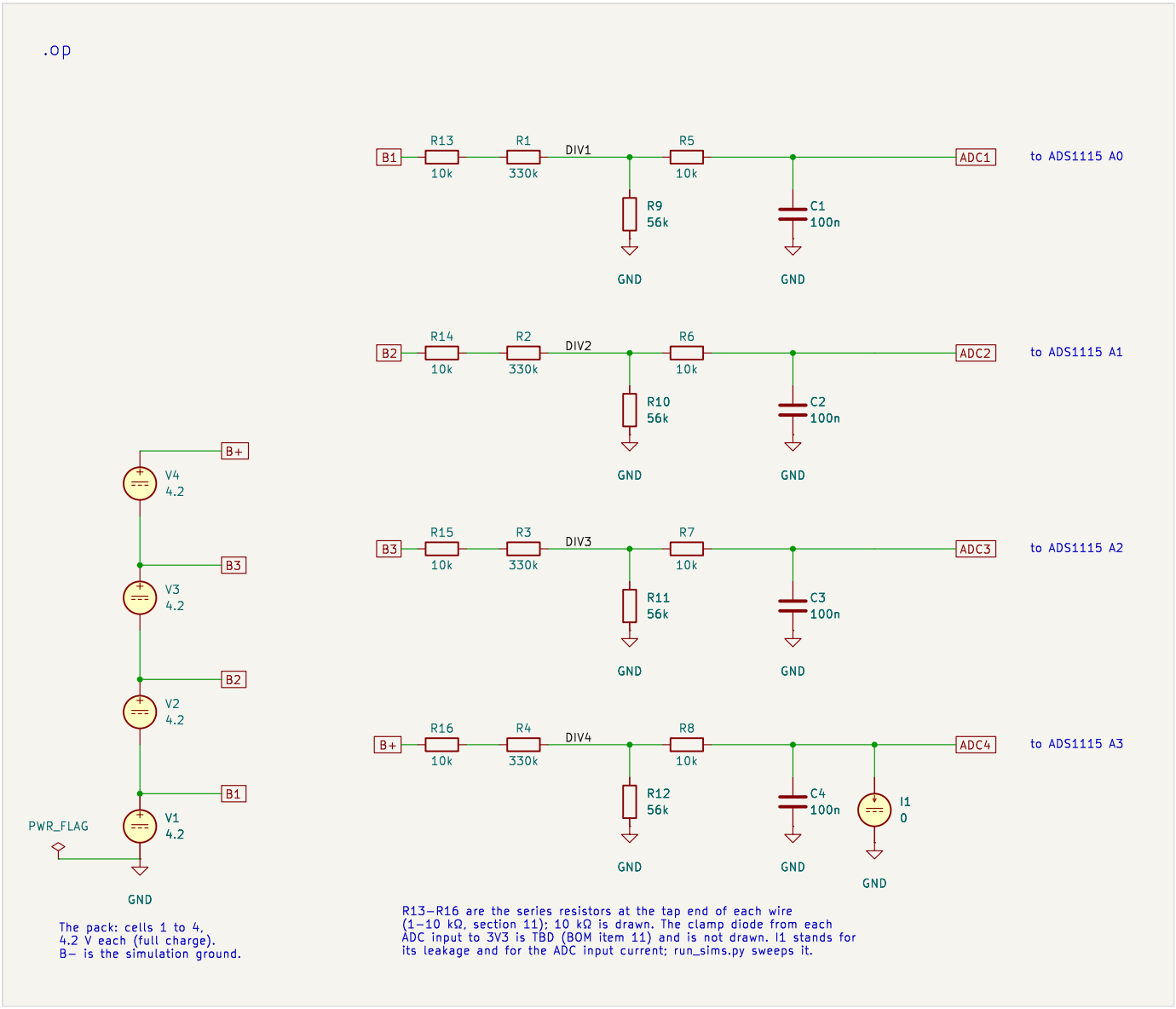


Figure 21. KiCad sheet of the four cell-tap channels. V1–V4 are the cells at full charge; R13–R16 are the tap resistors at the pack. I1 stands for leakage at an ADC input.

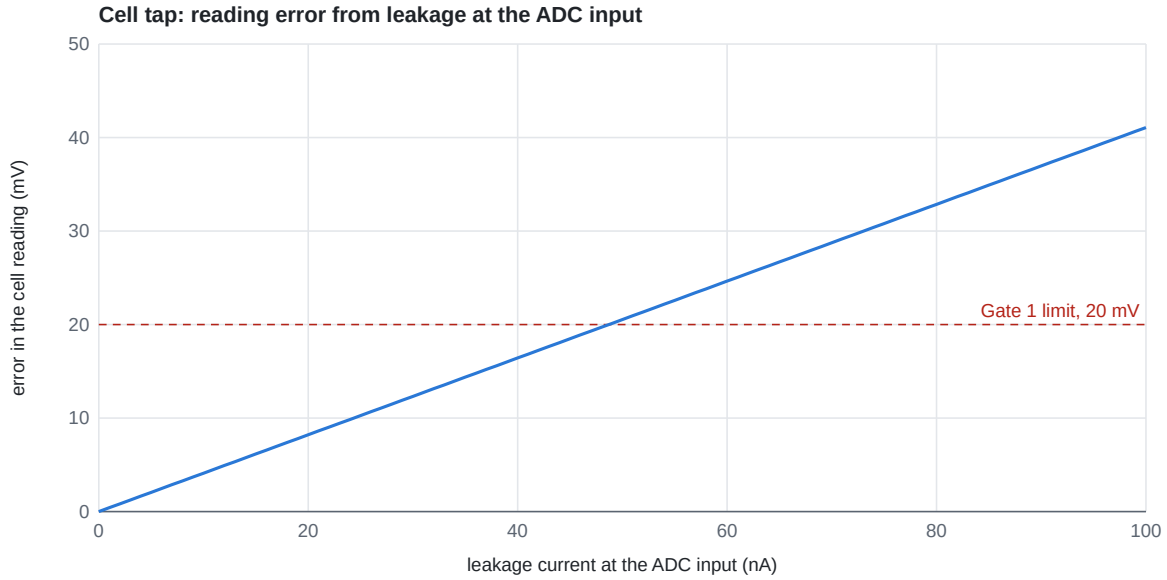


Figure 22. Leakage at an ADC input (from the clamp diode or the ADC itself) against the error in the cell reading. Calibration removes a steady leakage, but not one that changes with temperature, so the clamp must leak well below the 49 nA that would use the whole Gate 1 limit.

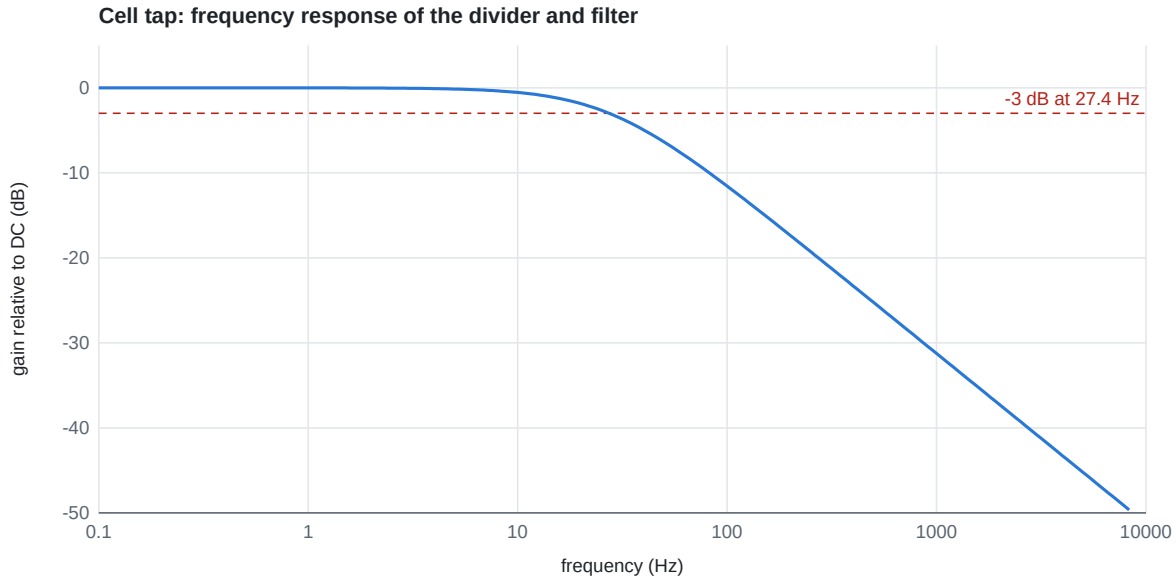


Figure 23. Frequency response from a tap to its ADC input. The filter passes the cell voltage and removes noise above about 27 Hz; after a step the input settles to 16-bit accuracy in 64 ms, less than one 200 ms sample.

16.2 NTC divider (board B)

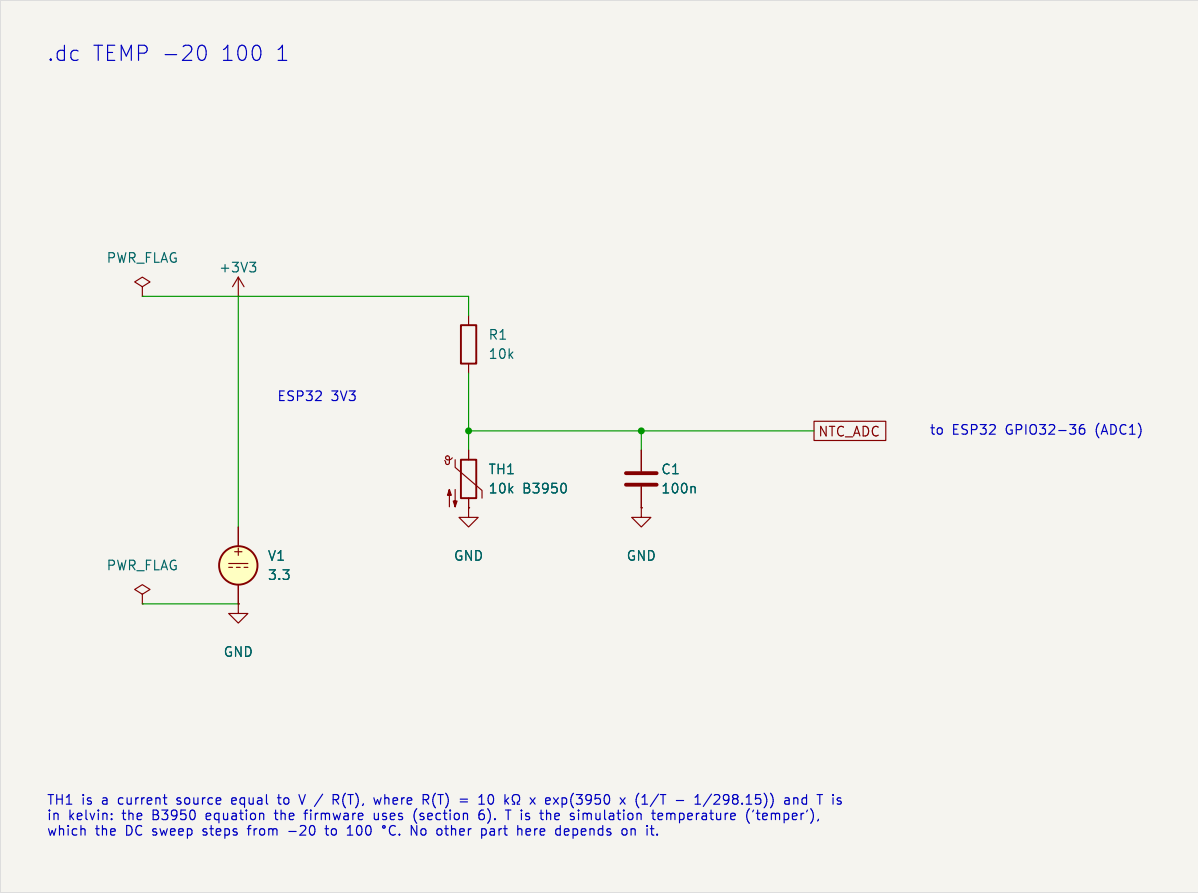


Figure 24. KiCad sheet of one NTC divider. TH1 follows the B3950 equation at the simulation temperature, which the DC sweep steps from −20 to 100 °C.

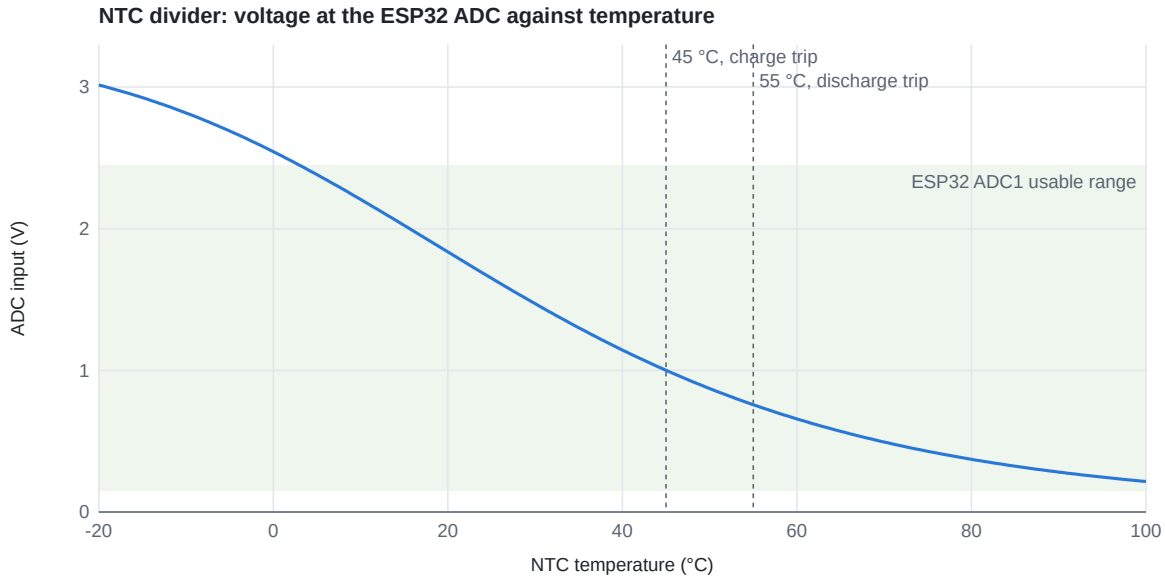


Figure 25. Voltage at the ESP32 ADC against the NTC's temperature. It stays in the ADC's accurate range from about 3 °C to above 100 °C; at the 45 °C and 55 °C trips it is 1.00 V and 0.76 V.

16.3 Current sense

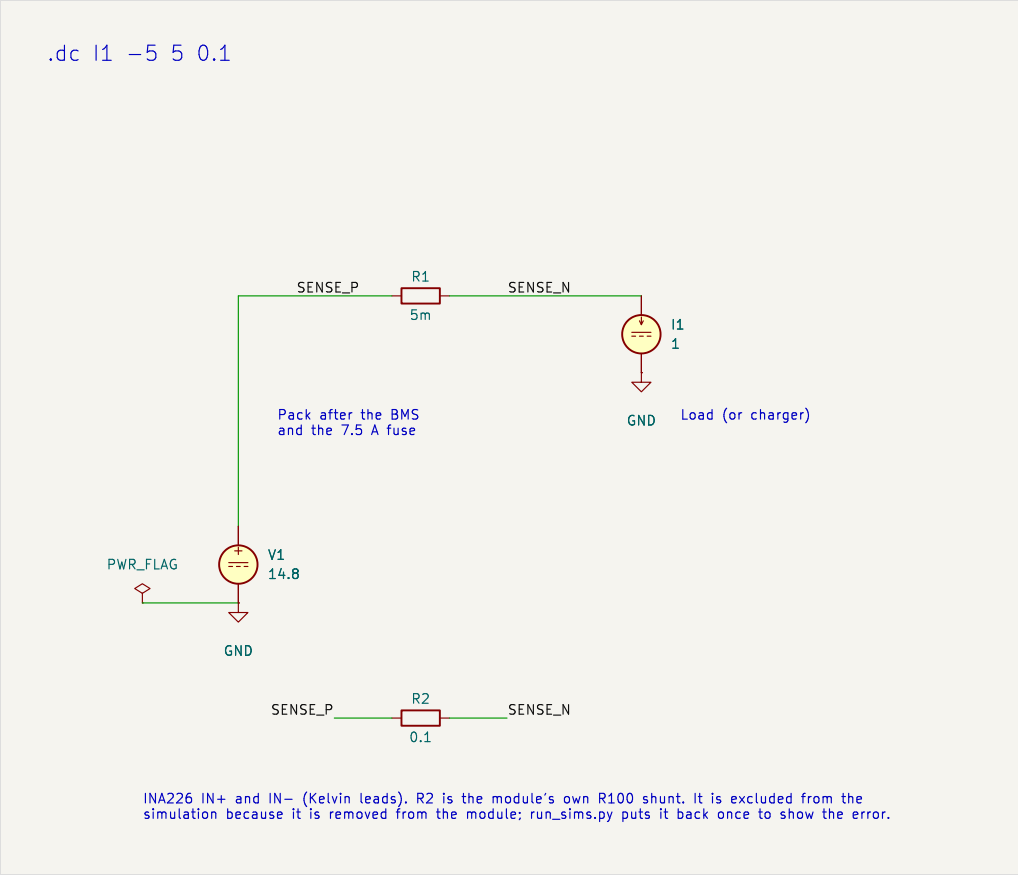


Figure 26. KiCad sheet of the 5 mΩ shunt and the INA226 inputs. R2 is the module's own R100 shunt, which is removed; it is excluded from the simulation.

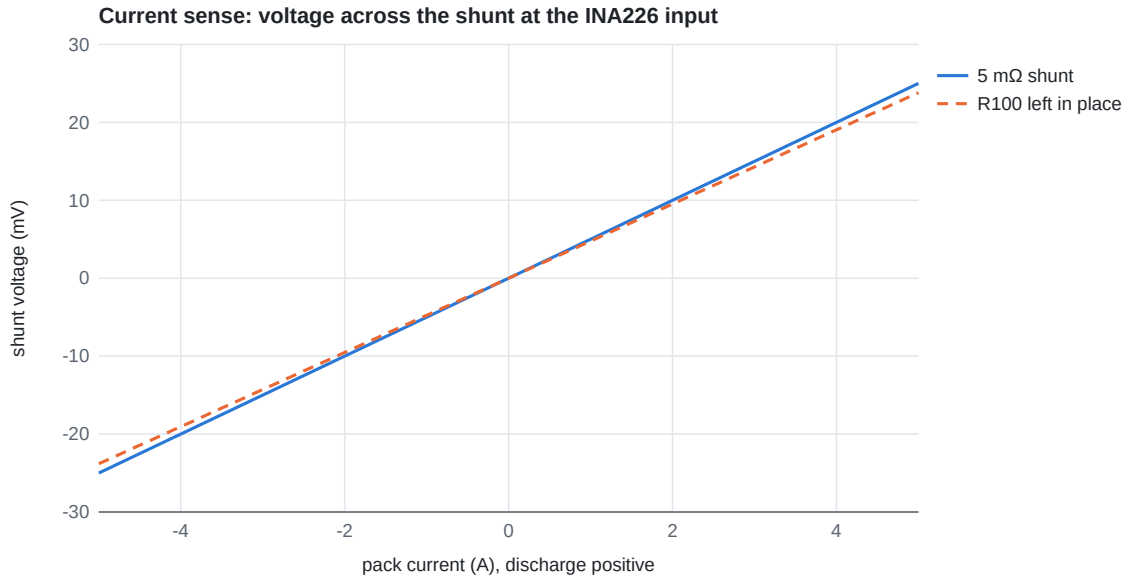


Figure 27. Shunt voltage from 5 A of charge to 5 A of discharge: 25.0 mV at 5 A, inside the INA226's ±81.92 mV. With R100 left in place the reading is 4.76 % low.

16.4 The thermal model of a cell

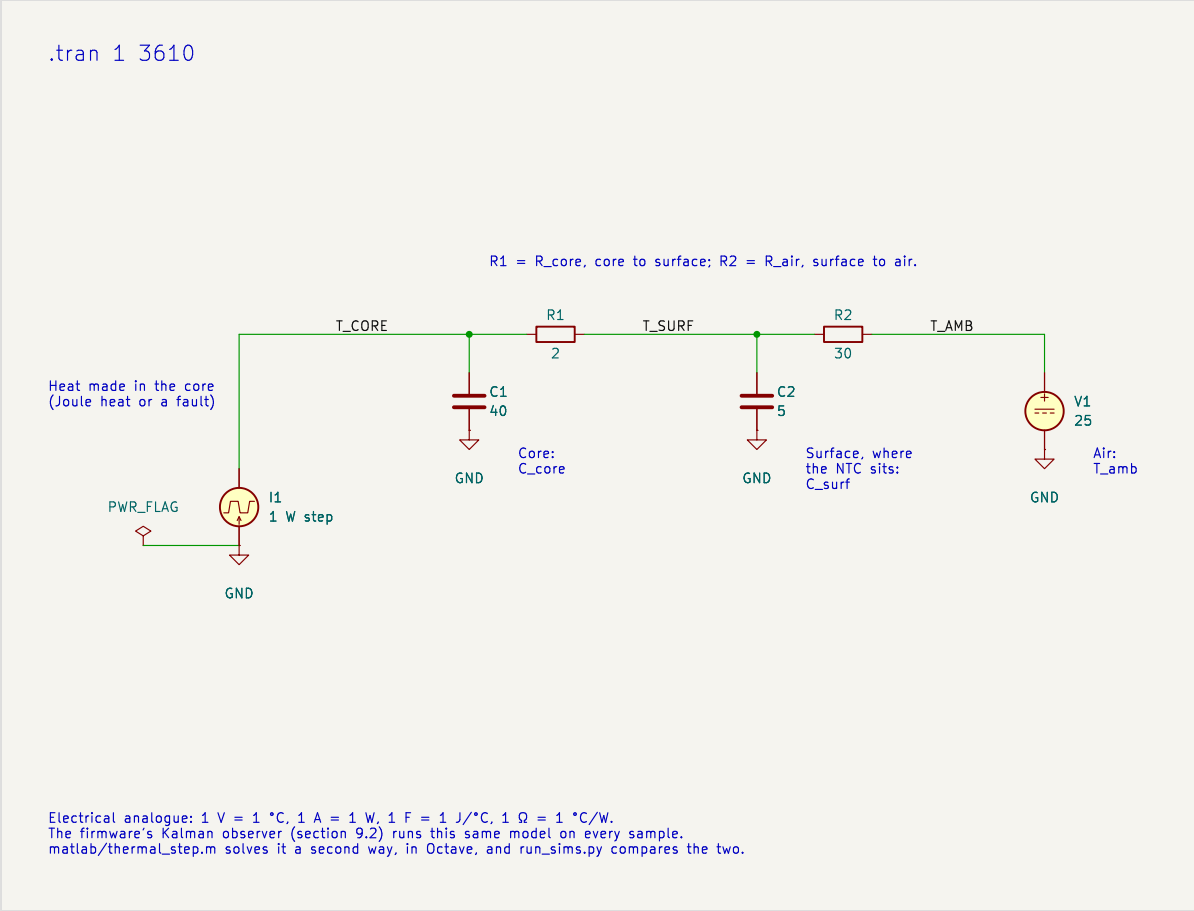


Figure 28. The two-node thermal model of section 9.1 as an electrical circuit: 1 V stands for 1 °C, 1 A for 1 W, 1 F for 1 J/°C and 1 Ω for 1 °C/W. I1 is 1 W of heat in the core from t = 10 s.

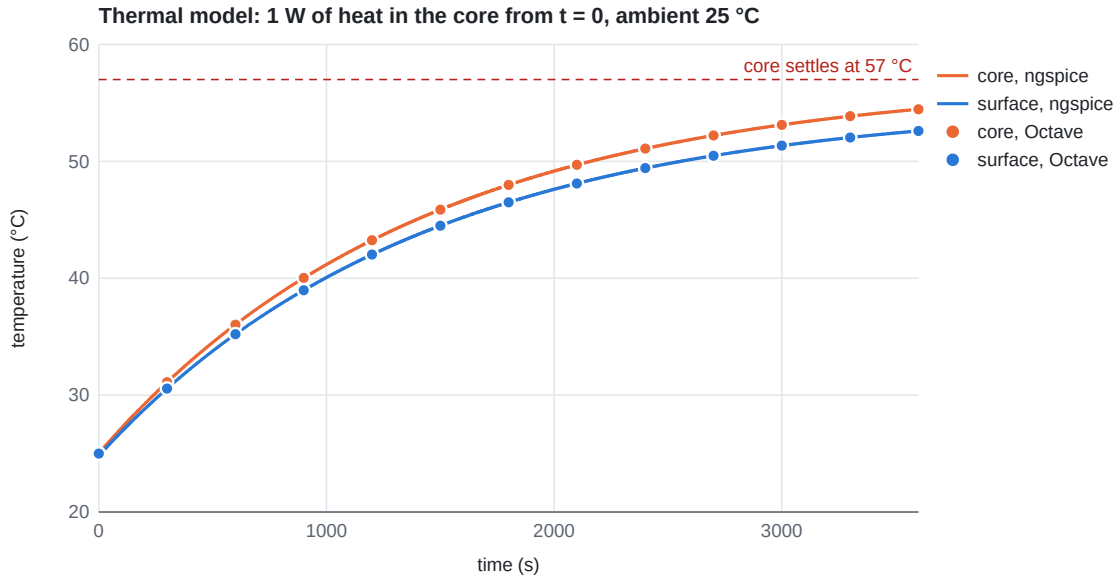


Figure 29. Core and surface temperatures after 1 W starts in the core. The lines are ngspice; the dots are the Octave script, which solves the same model a second way. The two agree to 0.0002 °C. The cell needs more than an hour to settle, which is why the supervisor estimates the heat instead of waiting for a fixed limit.

16.5 The observer in GNU Octave

`matlab/thermal_observer.m` is a copy of the firmware's Kalman observer (section 9.2), written in the `.m` language. `crosscheck` runs simulator scenarios with a log of every sample, gives the logged current and

temperatures to the Octave observer, and compares its core temperature and abnormal heat with the firmware's. The limits are 0.01 °C and 0.002 W; the differences come from the rounding of the log.

Scenario	Samples compared	Largest core-temperature difference	Largest abnormal-heat difference	Result
`discharge_steps`	24,600	0.0029 °C	0.0013 W	agree
`heat_no_current`	950	0.0066 °C	0.0013 W	agree
`load_defect`	4,307	0.0026 °C	0.0013 W	agree
`soft_short`	28,501	0.0040 °C	0.0010 W	agree

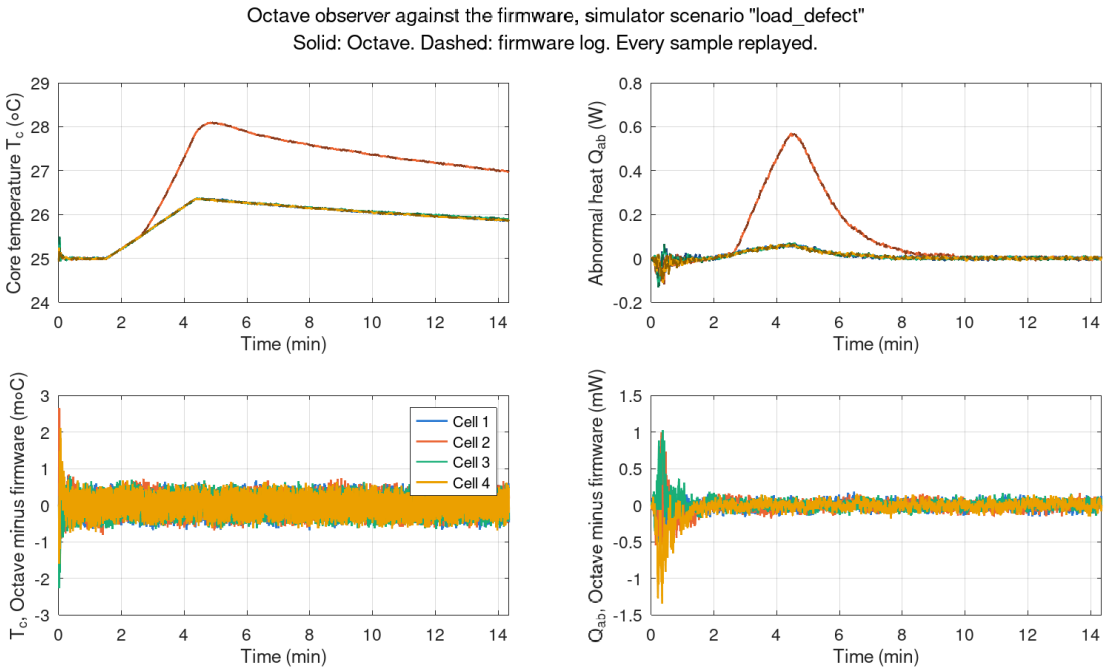


Figure 30. The Octave observer (solid) against the firmware (dashed) on the cell-defect scenario. Top: core temperature and abnormal heat of each cell. Bottom: the differences, in thousandths of a degree and thousandths of a watt.

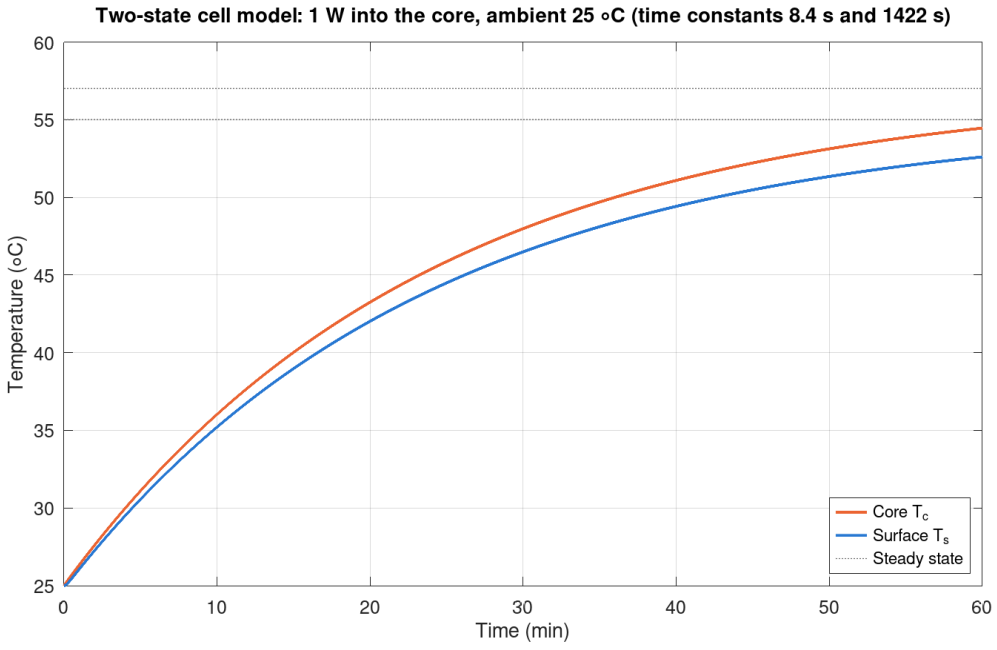


Figure 31. The step response drawn by Octave, with the steady-state temperatures.

16.6 The predictive layer in Scilab Xcos

Two block diagrams, in `scilab/`, open in Scilab 2027.0.0. The values are in each model's context (**Simulation → Set Context**): the firmware's development values. `cell_thermal_model.zcos` is the thermal model of section 9.1 as a state-space block; one hour after 1 W starts in the core, it agrees with ngspice and Octave within 0.004 °C.

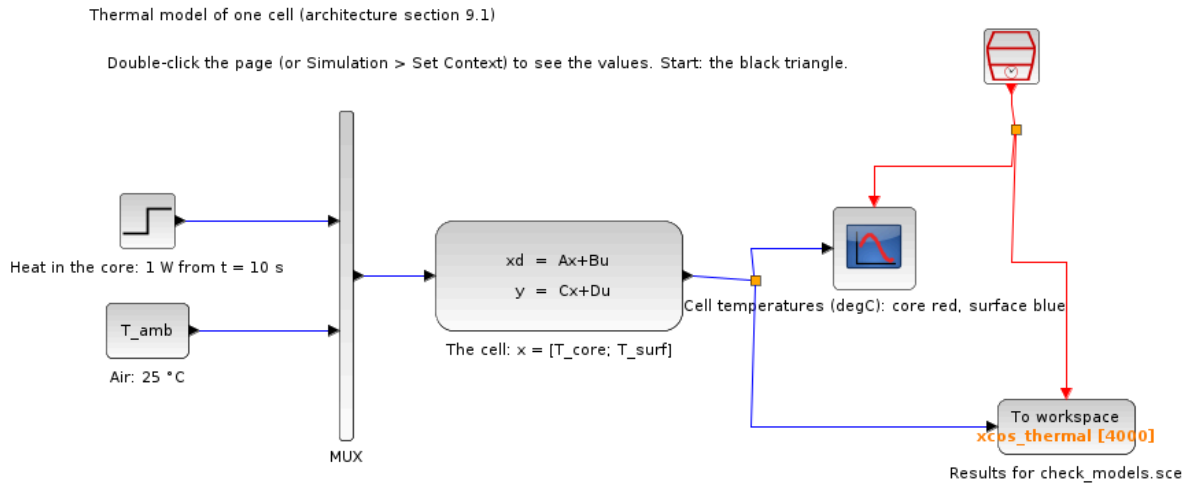


Figure 32. cell_thermal_model.zcos as Xcos draws it: the heat and the air temperature go into the cell's state-space model; the scope draws the core and surface temperatures.

`abnormal_heat_detection.zcos` gives a cell a 3 A load, whose Joule heat I²R the supervisor knows, and from minute 10 a further 1 W that it does not know: a fault. The observer is the firmware's Kalman filter with the gain it settles to, run every 0.2 s on the surface temperature alone. The predictive trip is set at an estimated abnormal heat of 0.75 W (1.0 °C/min, section 9.8); the comparison is a fixed 55 °C limit on the surface.

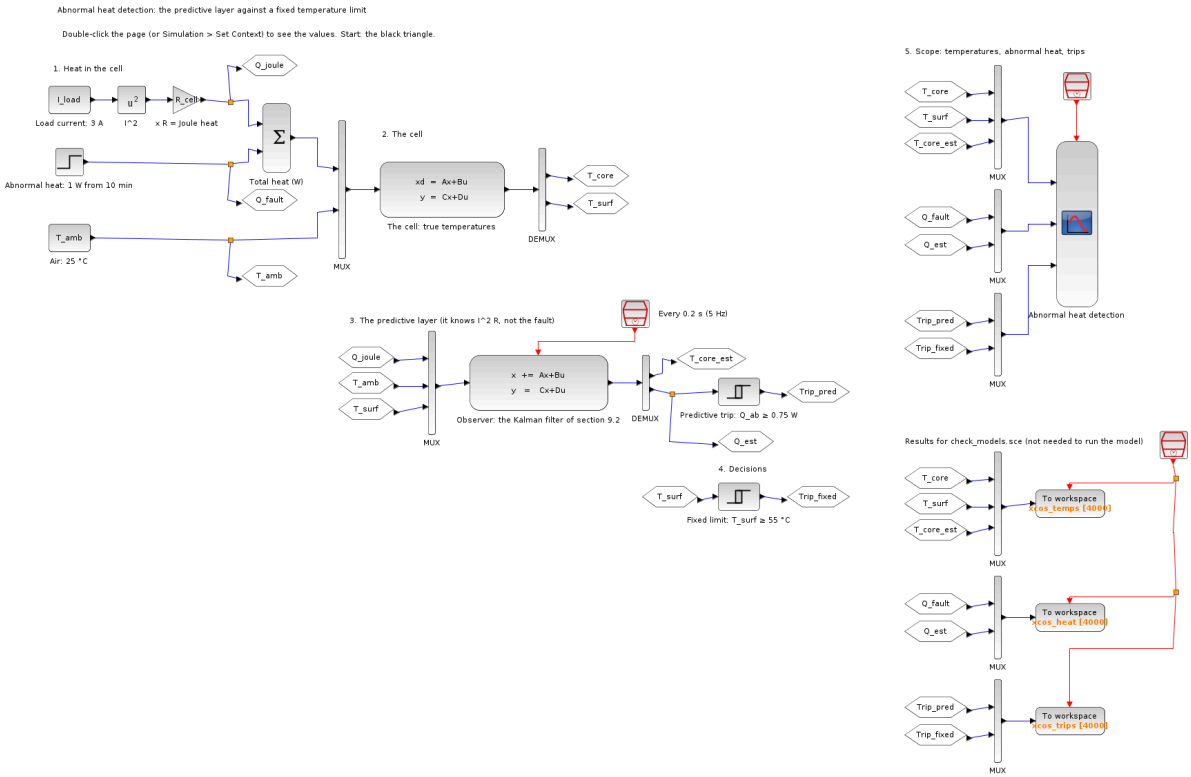


Figure 33. abnormal_heat_detection.zcos as Xcos draws it: 1. the heat, 2. the cell, 3. the observer, 4. the two trips, 5. the scope. Tags (Goto and From blocks) carry signals across the page.

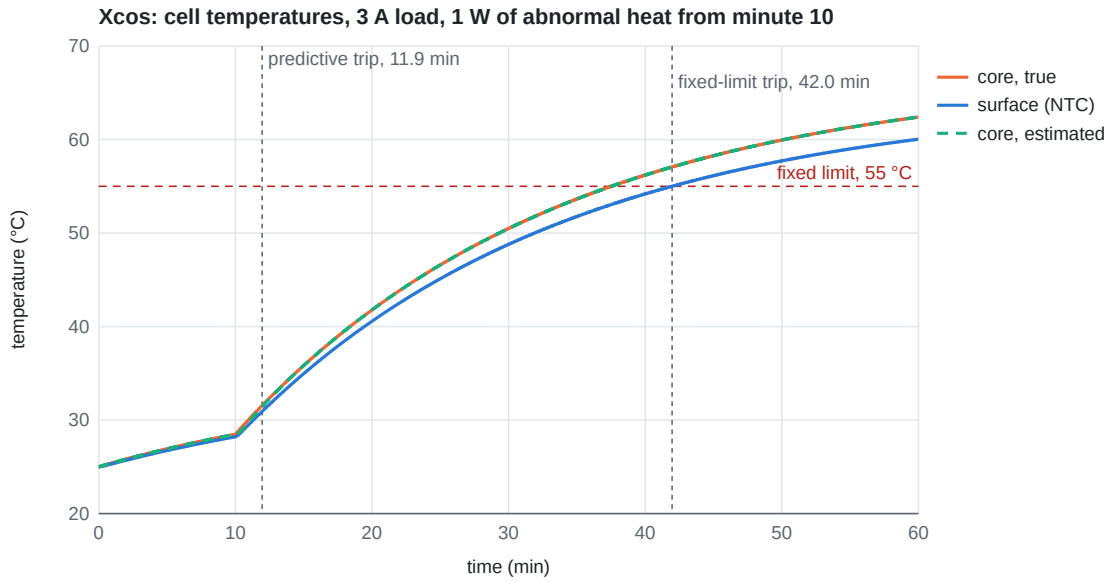


Figure 34. The cell's temperatures in the Xcos model. The predictive trip comes 117 s after the fault starts, with the surface at 30.9 °C; the fixed limit trips 32 minutes after it starts.

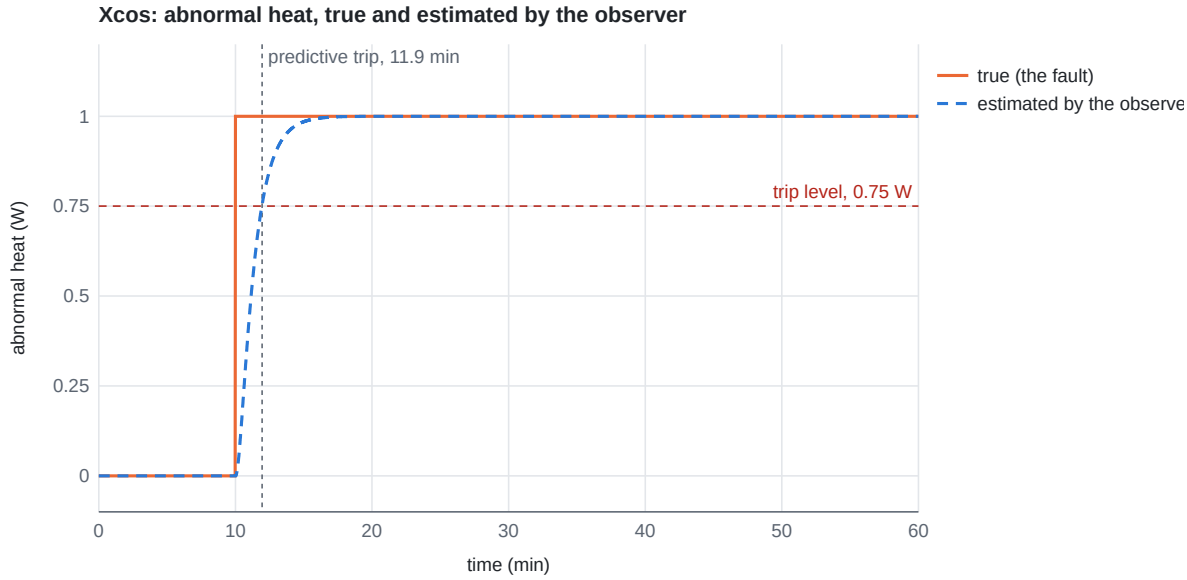


Figure 35. The abnormal heat: the true 1 W fault and the observer's estimate, which reaches the 0.75 W trip level in about two minutes and settles at 1.00 W. Before the fault it stays at zero. The model has no sensor noise, and the firmware also waits 10 s before it trips.

17. Safety rules

- No lithium cell is connected before Gate 3. The first live power-up and the first charge are never done alone.
- Never drive a live cell into thermal runaway, overcharge or over-discharge to prove a limit. Limits are proven with simulated signals, the simulator, a heated dummy, and the supervised 47 Ω resistor of Gate 11.
- Never solder to a cell terminal. Reject any cell that is swollen, dented, rusty, leaking, warm or smelling.
- Work on a non-conductive bench with eye protection, rings and watches off, every exposed terminal covered except the one being worked on, and a lithium-battery fire plan and sand bucket at hand.
- The hardware BMS stays connected as the final backstop; nothing in the firmware overrides it.
- The enclosure is vented and is not a fire enclosure: never seal it.